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Impact of polymer crosslinking on release mechanisms from long-acting levonorgestrel intrauterine systems

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Abstract

Polydimethylsiloxane (PDMS) crosslinking density is a critical material attribute of levonorgestrel intrauterine systems (LNG-IUSs) that affects drug release and may have a significant influence on product performance and safety. Accordingly, the objective of the present work was to investigate the impact of PDMS crosslinking on the release mechanisms of LNG-IUSs and thereby achieve better product understanding. To investigate the effect of PDMS crosslinking, LNG-IUSs with varying prepolymer ratios and different mixing conditions were prepared. Accelerated and real-time *in vitro* release of the LNG-IUSs were conducted for up to 80 days and 7 months, respectively. Contrary to conventional understanding, formulations with higher crosslinking density showed faster drug release rates. To further understand this anomalous release behavior, the microstructure and molecular properties (using scanning electron microscopy, mercury intrusion porosimetry, polymer swelling studies, solid-state silicon NMR, and wide-angle X-ray diffraction) were investigated. Interestingly, it was revealed that high PDMS crosslinking forms a solid-state porous branched network with amorphous polymer domains facilitating fast solvent uptake (in organic solvents) and easy access to the drug particles leading to rapid mass transport of the drug molecules. Furthermore, formulations processed using planetary mixing showed higher crosslinking densities and faster drug release rates than those prepared using

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CRediT authorship contribution statement

Suraj Fanse: Conceptualization, Methodology, Validation, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Quanying Bao:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Yuan Zou:** Writing – review & editing, Supervision, Project administration. **Yan Wang:** Writing – review & editing, Project administration. **Diane J. Burgess:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijpharm.2021.121383>.

manual mixing. Model fitting of all LNG-IUSs were carried out using first order, two-phase (zero order plus Higuchi), and Korsmeyer-Peppas models. The first order model (which showed the best fitting for the full release profile) was used to establish correlations between the drug release rates and the PDMS crosslinking densities of LNG-IUSs. This is the first comprehensive report providing novel insights into crosslinking-induced microstructural changes and physicochemical properties that dictate drug release from LNG-IUSs.

Keywords

Intrauterine device; Crosslinking density; Polydimethylsiloxane; Release mechanism; Drug-device combination; Silicone

1. Introduction

With increasing efforts to provide safe and effective treatment strategies, drug-device combination products are gaining more attention. Levonorgestrel intrauterine systems (LNG-IUSs) are complex drug-device combination products for controlled release which belong to the class of long-acting reversible contraceptives (LARCs). LNG-IUSs consist of a cylindrical drug-polymer reservoir (a mixture of LNG in a polydimethylsiloxane (PDMS) matrix) around the vertical stem of a T-shaped polyethylene frame. The drug-polymer reservoir is covered with an outer semi-opaque polymer membrane containing PDMS (Fig. 1A) (Mirena FDA product label, 2021). LNG-IUSs are designed to release the contraceptive drug for up to 7 years. LNG-IUSs involve non-surgical administration, offer higher patient compliance than the conventional forms of contraception, and provide a contraceptive efficacy of greater than 99.9% (Adeyemi-Fowode and Bercaw-Pratt, 2019; Beatty and Blumenthal, 2009; McNicholas and Peipert, 2012). Despite their unique advantages, very few products exist on the market. The U.S. Food and drug administration (FDA) has approved 4 LNG-IUSs since 2000: Mirena® (7 years), Liletta® (6 years), Kyleena® (5 years), and Skyla® (3 years) (FDA, 2021).

The complex nature of excipients, extremely long duration of release, and complicated release mechanisms are some of the challenges in the development of *in vitro-in vivo* correlations (IVIVCs) to predict the *in vivo* behavior of LNG-IUSs. Consequently, it is imperative to develop suitable release testing methods as well as investigate the critical quality attributes influencing the product performance and release mechanisms, to obtain a better understanding of this drug product.

PDMS is the release-controlling excipient present in both the matrix (core) and the outer membrane of LNG-IUSs. PDMS is a crosslinked polymer formed by the curing of silicone prepolymers or copolymers and consists of a repeating silicon and oxygen backbone (-Si-O-) in their structure (Colas and Curtis, 2004). The structural unit of PDMS is an important factor affecting PDMS crosslinking density. PDMS can exist in four types of monomer structures which are monofunctional (M), difunctional (D), trifunctional (T), and tetrafunctional (Q) (Colas and Curtis, 2004; Zeldin, 2003) (Fig. 1B). The arrangement of polymeric chain units and the length of the crosslinked chains are dependent on the monomer unit of PDMS (Kendrick et al., 1989). Addition curing, condensation curing,

and peroxide curing are the three methods of achieving PDMS crosslinking, with addition curing being the most commonly used. The commercially available PDMS polymers based on addition curing are supplied as two-part systems, containing prepolymers A and B. Prepolymer A (Part A) contains a vinyl-terminated PDMS elastomer base along with the catalyst, and prepolymer B (Part B) contains the crosslinking agent along with the elastomer base. The prepolymers also contain a small amount of other components such as hydrolytic products of silicones and additives (such as silica). PDMS crosslinking through addition curing involves a chemical reaction between the PDMS elastomer base (containing vinyl-terminated groups) and the crosslinking agent (containing methylhydrogensiloxane group) in the presence of a platinum-based catalyst (Mojsiewicz-Pieńkowska, 2015) (Fig. 1C).

Crosslinking leads to the formation of a hydrophobic network structure of PDMS which is required to achieve the controlled release of drug. PDMS crosslinking also imparts elasticity and mechanical strength to the network structure (Wang and Gallant, 2014). LNG-IUSs have a high drug loading to sustain drug release for several years. Crosslinking in PDMS is one of the key attributes of LNG-IUSs which may play an influential role on the drug release properties and thus product safety and efficacy. Drug release testing of qualitatively (Q1) and quantitatively (Q2) equivalent LNG-IUSs has been investigated by Bao et al. (2019). Fanse *et al.* have discussed the influence of the degree of PDMS crosslinking on the physicochemical and mechanical properties of LNG-IUSs (Fanse et al., 2021). However, there have been no previous studies investigating the effect of PDMS crosslinking density on the performance of LNG-IUSs and the mechanisms of controlled release achieved through PDMS remain unclear.

Therefore, the objective of the present work was to understand the impact of polymer crosslinking on the drug release mechanisms from LNG-IUSs. More specifically, following three factors were investigated: prepolymer ratios (B:A), LNG particle sizes, and mixing conditions. LNG-IUSs with different crosslinking densities were prepared using two approaches: 1) varying the PDMS prepolymer ratios; and 2) using different mixing methods during the processing of the LNG-IUSs. Two methods to determine the extent of PDMS crosslinking in LNG-IUSs were developed. The first method evaluated the degree of PDMS crosslinking (%) based on the soluble gel fraction of the LNG-IUSs using a solvent extraction technique. The second method utilized the polymer swelling properties to determine the PDMS crosslinking density (mol/cm^3) based on the Flory-Rehner theory (Flory, 1953). An accelerated release testing method was developed which provided complete release of LNG-IUSs within 2–3 months. Accelerated and real-time *in vitro* drug release testing of formulations with varying crosslinking densities was carried out and model fitting was performed to study the release kinetics. A correlation between the PDMS crosslinking density and the drug release rate was established. Scanning electron microscopy (SEM), mercury intrusion porosimetry, swelling studies, ^{29}Si nuclear magnetic resonance (NMR) and wide-angle X-ray diffraction were performed to investigate the microstructure. In addition, the effect of drug particle size on polymer crosslinking and drug release was evaluated. Drug release mechanisms based on the impact of polymer crosslinking have been proposed.

2. Materials and methods

2.1. Materials

Polydimethylsiloxane-based liquid silicone rubber supplied as two prepolymers (MED-4840 Part A and Part B) were purchased from Nusil™ (Carpinteria, CA, USA). Levonorgestrel was purchased from two sources: Tecoland (Irvine, CA, USA) (non-micronized API with a mean particle size of 16 µm) and Sterling chemical (Malta, Italy) (micronized API with a mean particle size of 3 µm). Tert-butanol (TBA, HPLC grade), acetonitrile (ACN, HPLC grade), tetrahydrofuran (THF, HPLC grade), toluene (HPLC grade), and hexane (HPLC grade) were purchased from Fischer Scientific (Fair Lawn, NJ, USA). Phosphate buffered saline (PBS, Bio performance certified, pH 7.4) and sodium chloride were purchased from Sigma-Aldrich (St. Louis, MO, USA). The PDMS silicone tubing (Silastic®) was purchased from Dow Corning (Midland, MI, USA). Unless otherwise specified, all other materials used were of analytical grade.

2.2. Preparation of levonorgestrel intrauterine systems

2.2.1. Formulation of drug-polymer reservoirs with different prepolymer ratios—A customized extrusion device comprising a twin-syringe assembly was used to prepare the drug-polymer reservoirs as described previously (Bao et al., 2018). Briefly, the twin syringe assembly was composed of a cylindrical polycarbonate mold, two 3-mL polypropylene syringes with the needle hub removed, two centering devices A and B (made of polycarbonate), and a metal rod (Fig. 2). Different stoichiometric ratios of prepolymers B:A (0.2:1, 0.25:1, 0.33:1, 0.5:1, 1:1, 2:1, 3:1, 4:1, and 5:1), and LNG (Tecoland) were mixed manually for 10 min to obtain nine different drug-polymer reservoirs. Each of the drug-polymer reservoirs were prepared with a batch size of 300 mg and a drug loading of 50% w/w LNG.

Drug-polymer reservoirs (with prepolymers B:A in a ratio of 1:1) were prepared using API of two different particle sizes (micronized and non-micronized) to investigate the effect of particle size on crosslinking. In addition, drug-polymer reservoirs (with prepolymers B:A in a ratio of 1:1) were prepared using different mixing methods (mixing using planetary mixer vs manual mixing) to investigate the effect of minor manufacturing difference on crosslinking. Drug-polymer reservoirs (with prepolymers B:A in a ratio of 1:1) were mixed using a planetary centrifugal mixer (DAC330–100 Pro Speedmixer®, Flacktek Inc., SC, USA) instead of manual mixing. The parameters of the planetary centrifugal mixer were as follows: 1500 rpm (10 s), 2500 rpm (20 s), 3500 rpm (4 min). The cycle was repeated twice and the total mixing time was 9 min.

After mixing, all of the drug-polymer reservoirs were dried under vacuum for 5 h. Each of the drug-polymer reservoirs were transferred into the syringe assembly of the extrusion device (Fig. 2B) and extruded to form hollow cylindrical drug-polymer reservoirs. These reservoirs were then cured at 80 °C for 20 h. The cured drug-polymer reservoirs were demolded and used for further fabrication.

2.2.2. Preparation of LNG-IUSs with different prepolymer ratios—The resultant drug-polymer reservoirs (with different prepolymer ratios, particle size, and mixing conditions) were cut into 100 mg pieces and enveloped onto nylon rods (diameter: 1.57 mm). The silastic® silicone tubing was cut into 24 mm long pieces and used as the outer PDMS membrane. The silicone tubing segments were allowed to swell in hexane for 5 min, and then coated onto the drug-polymer reservoirs in a symmetrical manner. The resultant LNG-IUSs were dried under vacuum overnight at room temperature to remove the residual hexane.

2.3. UPLC analytical method

Ultra performance liquid chromatography (UPLC) (Acquity H-class, Waters) method previously developed by Fanse *et al.* was used for quantification of the LNG-IUSs (Fanse et al., 2021). The mobile phase was a solution of 0.05% trifluoroacetic acid in a water:acetonitrile mixture (95:5) (Mobile phase A), and 0.05% trifluoroacetic acid in a water:acetonitrile mixture (5:95) (Mobile phase B). The mobile phase was pumped in a gradient condition at a flow rate of 0.4 mL/min for 5 min. A C18 BEH column (50 mm X 2.1 mm, 1.7 µm, Waters®) was used at 30 °C. The sample injection volume was 1 µL and the detection wavelength was 242 nm. The chromatograms were analyzed using Empower® 3.0 software.

2.4. Drug loading and content uniformity

The drug loading and content uniformity of the drug-polymer reservoirs (with different prepolymer ratios, particle size, and mixing conditions) were determined using a tetrahydrofuran extraction method (Fanse et al., 2021). Briefly, the drug-polymer reservoirs were cut into small pieces of approximately 12 mg from three different regions (to ensure content uniformity). Each of the pieces was accurately weighed and extracted in 25 mL of tetrahydrofuran in tightly sealed volumetric flasks for 48 h at room temperature. An appropriate volume of tetrahydrofuran was then added to achieve a final volume of 25 mL. The resultant samples were vortexed for 2–3 min, diluted with a mixture of acetonitrile and water (60:40), and injected into the UPLC for analysis.

2.5. Determination of polymer crosslinking in LNG-IUSs

2.5.1. Degree of PDMS crosslinking determined using solvent extraction (Method-I)—A solvent swelling and extraction method was used to determine the degree of polymer crosslinking achieved in the LNG-IUSs as described previously (Fanse et al., 2021) (Fig. 3). Briefly, approximately 12 mg of each of the drug-polymer reservoirs were extracted using 25 mL of tetrahydrofuran for 48 h. The amount of LNG extracted was quantified using UPLC. The reservoirs were then vacuum dried for 20 h at room temperature. The weights of the reservoirs before and after drying were recorded. Further, the extraction process was repeated using 20 mL of hexane for 48 h. The reservoirs were separated from hexane and dried under vacuum for approximately 18 h. The differences in their weights before and after drying were recorded. The degree of PDMS crosslinking (% of total weight) was determined from the difference in the total non-crosslinked material extracted during the entire process and the initial weight of the drug-polymer reservoir used. Each of the

drug-polymer reservoirs were analyzed in triplicate. The degree of PDMS crosslinking in LNG-IUSs was calculated based on equation (i).

$$\text{Degree of crosslinking (\%)} = \left[\frac{W_0 - (\text{Noncrosslinked material in THF} + \text{Noncrosslinked material in hexane})}{W_0} \right] \times 100 \quad (1)$$

where,

Non-crosslinked material in tetrahydrofuran = $\Delta W_{THF} - W_{LNG}$

ΔW_{THF} (total weight loss in tetrahydrofuran) = $W_0 - W_1$

W_{LNG} = amount of LNG extracted in tetrahydrofuran (determined by UPLC) (mg)

W_1 = weight of the reservoir after vacuum drying following extraction in tetrahydrofuran (mg)

W_0 = initial weight of drug-polymer reservoir used (mg)

Non-crosslinked material in hexane = $W_1 - W_2$

W_2 = weight of the reservoir after vacuum drying following complete extraction in hexane (mg)

2.5.2. PDMS crosslinking density determined using solvent swelling

(Method-II)—Approximately 12 mg of each of the drug-polymer reservoirs were soaked in 25 mL of toluene and allowed to swell for 5 days. The extraction time was optimized during method development to ensure complete extraction of the non-crosslinked polymer. The equilibrium-swollen drug-polymer reservoirs were removed from the solvent and their weights were recorded. The reservoirs were then dried under vacuum for about 16 h at room temperature and the dried weights were recorded. The polymer volume fractions in the equilibrium-swollen drug-polymer reservoirs were determined from the swelling experiments. The densities of PDMS with different mixing ratios of prepolymers were determined using a pycnometer method. The Flory-Huggins solvent-polymer interaction term for PDMS-toluene was 0.465 and the molar volume of toluene was 106.85 cm³/mol (Bueche, 1955; Smallwood, 1996). The crosslinking density is defined as the number of crosslinks per unit volume in a polymer network. The Flory-Rehner equation as described below (equation ii) was used to calculate the crosslinking density (mol/cm³) in LNG-IUSs with different prepolymer ratios (Flory, 1953).

$$n = - \frac{[\ln(1 - v_2) + v_2 + \chi_1 v_2^2]}{V_1(v_2^{1/3} - v_2/2)} \quad (2)$$

where, n = number of crosslinks per unit volume (mol/cm³)

v_2 = volume fraction of polymer in the swollen mass,

χ_1 = Flory-Huggins solvent-polymer interaction term,

V_1 = molar volume of the solvent

2.6. Particle size

The particle size of LNG used to prepare the drug-polymer reservoirs of LNG-IUSs was determined using a Malvern Mastersizer® 3000E laser diffractometer (Malvern Panalytical Ltd., UK) equipped with a wet sampling system accessory (HydroMV). Approximately 20 mg of LNG was dispersed in 10 mL of 0.5% w/v polyvinyl alcohol in water. The sample was vortexed for 1 min and sonicated for 20–30 s prior to particle size determination. The sample was added to the media (Milli-Q water) stirred at 3000 rpm until a laser obscuration of 12–15% was achieved. The volume-weighted particle size distribution was analyzed and all measurements were performed in triplicate.

2.7. Wide angle X-ray scattering (WAXS)

The structural characteristics of the drug-polymer reservoirs with different crosslinking densities were studied using wide angle X-ray scattering (WAXS). An Oxford wide angle diffractometer (Oxford Diffraction XCalibur PX Ultra) equipped with an Onyx CryoJet coldstream and Onyx CCD 2D detector was used to acquire the diffraction patterns at Cu-K α radiations generated at 35 kV and 35 mA. The phi-scan width was set to 0.15 and the exposure time was 200 s. Azimuthal integration around the beam center was performed to obtain the one-dimensional (1D) WAXS pattern. All experiments were performed at room temperature. The data were analyzed using the CrysAlis Pro software.

2.8. Solid state ^{29}Si NMR

Solid-state ^{29}Si CP/MAS (cross polarization/magic angle spinning) NMR analysis was performed using a Bruker Avance III 400 MHz spectrometer (Bruker, USA) with a 4 mm MAS probe and a resonance frequency of 79.49 MHz for ^{29}Si . Each of the drug-polymer reservoirs were cut into small pieces and placed in a zirconia rotor. The spectra were collected at a spinning speed of 5000 Hz, 4 μs of 90° pulses, a relaxation delay of 8 s, and a temperature of 25 °C. The chemical shifts of the samples were referenced relative to the external tetramethylsilane (TMS) standard. Data analysis was performed using MRestNova software (Mestrelab research, S.L, Spain).

2.9. In vitro drug release of LNG-IUSs

2.9.1. Accelerated in vitro drug release—Accelerated *in vitro* drug release of LNG-IUSs were carried out using a water shaker bath agitated at 100 rpm for up to 80 days. The release media was 200 mL of 45% v/v *tert*-butanol in pH 7.4 PBS at a temperature of 65 °C. The release test was conducted in glass bottles (PYREX®, Corning Inc.) with screw caps (GL45). Two small steel clips were clamped on the nylon rod of the LNG-IUSs to ensure that they sink to the bottom of the glass bottles. Sink conditions (10 times the saturation solubility of LNG in *tert*-butanol) were maintained throughout the experiments. The sampling plan for the accelerated release testing was as follows: 1 mL samples were

withdrawn at 15 min, 30 min, 1 h, 2 h, 3 h, 4 h, 5 h, 6 h, 8 h, 10 h, 12 h and 24 h on day 1. The samples were replenished with fresh media after every sampling. 1 mL samples were withdrawn daily up to 1 week and replenished with fresh media. Following the day 7 sampling, all the media in the bottles were drained and replenished with fresh release media. Thereafter, 1 mL samples were withdrawn every subsequent fourth and seventh day of the week and replenished with fresh release media. A full media exchange was performed every two weeks. All the experiments were performed in triplicate. The concentrations of LNG in the samples were determined using UPLC method.

2.9.2. Real-time in vitro drug release—The real-time *in vitro* release testing of LNG-IUSs were performed in 300 mL of 0.9% w/v NaCl at 37 °C for up to 7 months. 1 mL samples were withdrawn on days 1, 2, 3, 4 and 7 in the first week, and replenished with fresh release media. Following the day 7 sampling, all the media in the bottles were drained and replenished with fresh media. Thereafter, 1 mL samples were withdrawn every week and all the media in the bottles were replenished with fresh media following sampling. All the experiments were performed in triplicate. The concentrations of LNG in the samples were determined using UPLC method.

2.10. Morphology

The drug distribution and the morphology of the drug-polymer reservoirs (with different prepolymer ratios, particle sizes, and mixing conditions) were determined using a scanning electron microscope (FEI Nova NanoSEM 450 unit). Thin cross sections of the samples were mounted on carbon taped aluminum stubs and sputter coated with gold with a deposition rate of 6 nm/min for 3 min at 6 mA before imaging.

2.11. Swelling ratio

Swelling of the drug-polymer reservoirs with different crosslinking densities were conducted in *tert*-butanol at room temperature. The drug-polymer reservoirs were cut into 2 cm lengths and soaked in 50 mL of *tert*-butanol for 1 week. The solvent was replenished every two days. The reservoirs were then withdrawn from the media and the excess *tert*-butanol on the surface was removed. The length of the drug-polymer reservoirs was measured before and after the swelling test using a caliper (0.1 mm accuracy). All swelling experiments were performed in triplicate. The swelling ratio of the reservoirs in *tert*-butanol was calculated using the following equation:

$$\text{Swelling ratio} = L_{eq}/L_0 \quad (3)$$

where, L_{eq} represents the length of the drug-polymer reservoir at equilibrium, and L_0 represents the initial length.

2.12. Porosity

The porosity of the drug-polymer reservoirs of LNG-IUSs with different crosslinking densities were determined using a mercury intrusion porosimeter (AutoPore IV 9500, Micromeritics). Approximately 1 g of drug-polymer reservoirs were introduced into the

porosimeter and tested at a mercury filling pressure of 0.53 psi. The total intrusion volume, total pore area and porosity (%) of the reservoirs were recorded.

$$[\text{Porosity (\%)} = (1 - \text{bulk density/apparent particle density}) \times 100]$$

2.13. Statistical analysis

The data analysis, linear regression and data fitting were performed using Graphpad Prism software (Graphpad software, San Diego, CA, USA) and Excel 2016 (Microsoft Corporation, Redmond, WA, USA). All data were presented as average $\pm$ standard deviation (SD).

3. Results

3.1. Drug loading and content uniformity

The drug loading and content uniformity of the drug-polymer reservoirs (with different prepolymer ratios, particle sizes, and mixing conditions) were determined using the tetrahydrofuran extraction method. The target drug loading of all the drug-polymer reservoirs was 50% w/w LNG. The %RSD of the triplicate samples tested from different regions of each of the drug-polymer reservoirs were lower than 3% indicating satisfactory content uniformity. The drug loading varied with the prepolymer ratios with the highest drug loading was observed for the formulation with a prepolymer B:A ratio of 1:1 (Table 1). The observed drug loading decreased as the amount of either prepolymer A or prepolymer B increased.

At a constant prepolymer ratio of 1:1, planetary speed mixing and manual mixing of the drug-polymer reservoirs were compared. It was observed that the drug loading was slightly lower for the formulations prepared *via* planetary speed mixing. The drug-polymer reservoirs prepared using different particle sizes did not show significant difference in their drug loading.

3.2. Determination of PDMS crosslinking in LNG-IUSs

The extent of PDMS crosslinking in LNG-IUSs were determined using two different methods. Method-I was based on the extractable fraction of the non-crosslinked prepolymer in tetrahydrofuran and hexane, which measures the degree of PDMS crosslinking (expressed as % of total weight). Tetrahydrofuran and hexane were used in this method as they are strong solvents for the drug and the prepolymers, respectively. Method-II was based on the Flory-Rehner theory to determine the PDMS crosslinking density (mol/cm^3) which is a measure of the number of polymer chain segments bound by the crosslinks on both ends in unit volume. The lowest crosslinking density was observed at a prepolymer B:A ratio of 0.2:1. As the prepolymer B:A ratio is increased, the crosslinking density also increased up to the prepolymer B:A ratio of 1:1. Further increase in the ratio of the prepolymers B:A resulted in a reduction in the PDMS crosslinking density. The results indicated that the formulations follow the same rank order of PDMS crosslinking densities (Fig. 4A and B).

The crosslinking densities of the drug-polymer reservoirs with the prepolymer B:A ratio of 1:1 did not show significant differences with change in the drug particle size (p-value > 0.05). However, the crosslinking densities of the drug-polymer reservoirs prepared using the planetary speed mixer were slightly higher than those prepared using manual mixing (Fig. 4C).

3.3. Accelerated and real-time *in vitro* drug release from LNG-IUSs

The drug release from LNG-IUSs under real-time conditions is extremely slow. For example, the cumulative drug release from commercial LNG-IUSs at the end of 5 years is about 60% for Mirena® and 49.4% for Liletta (Creinin et al., 2016; Liletta FDA product label, 2019; Mirena FDA product label, 2021). It is reported that the initial daily drug release from the commercial product Mirena® is approximately 20 µg/day and complete drug release is not achieved over the duration of use for this product (Mirena FDA product label, 2021). Accordingly, a suitable accelerated release testing method with good discriminatory ability was developed to provide understanding of the release mechanisms involved.

Accelerated drug release of LNG-IUSs was conducted using 45% v/v *tert*-butanol in PBS at 65 °C in a water shaker bath stirred at 100 rpm. All the formulations with different crosslinking densities showed complete release within 80 days. As shown in Fig. 5A, the slowest drug release was observed for the formulation with the prepolymer B:A ratio of 0.2:1. The drug release rate increased as the ratios of the prepolymers B:A were increased up to 1:1. The formulation with the prepolymer B:A ratio of 1:1 demonstrated the highest drug release. A decline in the drug release was observed as the ratios of the prepolymers B:A were further increased from 1:1 up to 5:1. It was also observed that the formulations with very low crosslinking densities (prepolymer B:A ratios of 0.2:1, 0.25:1 and 5:1) showed burst release (for up to 6 h) since the drug was not completely entrapped in the polymer matrix (Fig. 5B).

The real-time *in vitro* drug release of LNG-IUSs with different prepolymer ratios at 37 °C in saline using a water shaker bath at 100 rpm is shown in Fig. 5C. The real-time drug release studies reported here are for up to 7 months and are used for comparison with the accelerated drug release data. The formulations with prepolymer B:A ratios of 5:1 and 0.2:1 (low crosslinking densities) initially showed faster drug release rates as compared to the formulation with a prepolymer B:A ratio of 1:1. These results are in agreement with the burst release of the same formulations observed under the accelerated testing conditions. The daily drug release rate of the formulations with low crosslinking densities drastically reduced with time. After approximately 170 days of drug release, the formulation with the prepolymer B:A ratio of 1:1 (highest crosslinking density) showed the highest drug release rate (Fig. 5C). The real-time drug release of LNG-IUSs (for 7 months) followed the same rank order as observed under the accelerated conditions.

The drug-polymer reservoirs with a prepolymer B:A ratio of 1:1 prepared using different mixing conditions showed differences in their accelerated *in vitro* drug release profiles. Accelerated release of the formulations prepared using planetary speed mixing showed faster drug release than those prepared using manual mixing (Fig. 6A and B). On the other hand, the real-time release profiles of these formulations did not show any significant

differences (Supplementary data). This is probably a result of the real-time release of LNG-IUSs being extremely slow and only approximately 2.5% cumulative drug release was observed for up to 4 months. The accelerated release profiles started to show differences in release for the formulations prepared using various mixing methods after 20% of drug release had occurred. Based on the release rate of these formulations under real-time conditions, approximately 20% of drug release would be obtained at 2.5 years and thus any difference in real-time release may be expected after 2.5 years.

The effect of particle size on drug release was investigated by preparing drug-polymer reservoirs using LNGs with different particle size at a constant prepolymer B:A ratio (1:1) and at both mixing conditions (manual and planetary speed mixing). The particle size (volume-weighted distribution, D_{90}) of LNG obtained from Tecoland (non-micronized) was determined to be $10.70 \pm 0.20 \mu\text{m}$ (span value of 7.89 ± 0.24), and the particle size of LNG obtained from Sterling (micronized) was determined to be $5.05 \pm 0.04 \mu\text{m}$ (span value of 3.43 ± 0.03) using the Malvern particle size analyzer. The accelerated release of LNG-IUSs prepared with micronized and non-micronized LNG, using both manual mixing and planetary mixing are shown in Fig. 6C and D. Surprisingly, it was observed that the drug particle size did not have any significant impact on drug release from the LNG-IUSs.

The real-time release for LNG-IUSs with different drug particle size is a six year study and accordingly the data for up to 4 months is shown (Supplementary data). The real-time release of formulations prepared using micronized and non-micronized LNG showed that particle size does not have any significant impact on the release behavior of LNG-IUSs. The results of real-time *in vitro* release are consistent with the observations of the accelerated release.

3.4. Release kinetics modeling of LNG-IUSs with different degrees of crosslinking

Since real-time release of LNG-IUSs is extremely slow, modelling was conducted using the accelerated release data. A previous report on Q1/Q2 equivalent LNG-IUSs had shown comparable results for the first order model and the two-phase model for drug release (Bao et al., 2019). However, the release kinetics may vary significantly as the PDMS crosslinking density in LNG-IUSs is changed (non-Q1/Q2 LNG-IUSs using different prepolymer ratios or mixing conditions). Release kinetics modeling of the LNG-IUSs were performed using the zero order, Higuchi, two-phase (zero order plus Higuchi), first order, and the Korsmeyer-Peppas models. The zero order model and the Higuchi model for the full release profiles of the LNG-IUSs showed poor data fitting (data not shown). Good model fitting were observed for: 1) the two-phase model (zero plus Higuchi); 2) the Korsmeyer-Peppas model; and 3) the first order model. The fitting parameters are shown in Tables 2A and 2B.

Analysis of the two-phase model revealed that the initial release (zero to approximately 30%) showed a good fit for zero order kinetics. The release data from approximately 30% to complete release of all LNG-IUSs investigated here followed the Higuchi model. The Korsmeyer-Peppas model showed good fitting for drug release up to 60%. However, the first order model demonstrated the best fitting for the complete release profiles (0–100% release) of the LNG-IUSs investigated here. The release rates obtained from the first order model were used for data comparison and predicting the release from LNG-IUSs with

different crosslinking densities. The time required to predict different percentages of drug release (*e.g.*, $T_{50\%}$) could be obtained from the fitting parameters. A relationship between the PDMS crosslinking densities and the first order release rate constants indicated that the release rate is directly proportional to the PDMS crosslinking density in LNG-IUSs (Fig. 7A). A linear correlation was established for all the formulations except the formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, and 0.33:1 (Fig. 7B).

3.5. Effect of molecular weight between the crosslinks (Mc) on drug release

Differences in the PDMS crosslinking densities due to variation in the prepolymer ratios (elastomer base and the crosslinking agent) may induce changes in the polymer chain length and orientation of the polymer chains connecting the adjacent crosslink junctions (Peppas et al., 2006). Thus, the molecular weight of the polymer chains between the adjacent crosslinks may vary. The average molecular weights of the PDMS chain segments between the crosslinks (denoted as M_c) were determined using the Flory-Rehner equation. The parameter M_c is calculated using the ratio of the density of polymer (ρ) (g/cm^3) to its crosslinking density (n) in mol/cm^3 (equation iv) (Flory, 1953; Peppas and Merrill, 1977).

$$M_c(\text{g/mol}) = \frac{\rho \left(\text{g}/\text{cm}^3 \right)}{n \left(\text{mol}/\text{cm}^3 \right)} \quad (4)$$

It was observed that M_c was the highest at a prepolymer B:A ratio of 0.2:1. M_c decreased as the prepolymer B:A ratio was increased up to 1:1. Further increase in the prepolymer B:A ratio up to 5:1 led to an increase in the M_c (Fig. 8A). A correlation between the molecular weight (M_c) and crosslinking density (n) revealed that the molecular weight between the two crosslink junctions decreases as the crosslinking density increases (Fig. 8B). The relationship between drug release rates of the LNG-IUSs and M_c indicated that higher drug release is observed for the formulations with lower M_c (Fig. 8C).

3.6. Effect of crosslinking on crystallinity in LNG-IUSs

To assess the impact of crosslinking on crystallinity of the drug and polymer, wide angle X-ray scattering of drug-polymer reservoirs with different prepolymer ratios were performed. The drug-polymer reservoirs with different crosslinking densities showed differences in their X-ray diffraction intensities. 2D WAXS frames showed a distinct pattern of Debye rings indicating a high overall crystallinity in all the formulations (Fig. 9A). It was observed that the formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, 0.33:1, and 5:1 showed greater diffraction intensities. The 1D WAXS diffractograms exhibit sharp peaks of high intensities between 13° and 35° (2θ) characteristic of the crystalline structure of LNG are superimposed over a broad halo pattern that is attributed to the amorphous PDMS (Fig. 9B). The formulation with a prepolymer B:A ratio of 1:1 exhibited the lowest intensity of crystalline peaks. A gradual increase in intensities of the peaks were observed as the prepolymer B:A ratios decreased from 1 to 0.2. Similarly, an increase in the peak intensities were seen when the prepolymer B:A ratios increased from 1 to 5. The representative X-ray diffractogram of pure drug (LNG) is shown in the Supplementary data.

The contributions of the amorphous and the crystalline domains were deconvoluted by profile fitting and the crystallinity index (expressed as % crystallinity) was calculated using the following equation (Sami et al., 2010).

$$\text{Crystallinity(\%)} = \frac{A_c}{A_c + A_a} \quad (5)$$

where, A_c = area of the crystalline peaks, and $A_c + A_a$ = total area obtained from the fitting (area of crystalline and amorphous domains).

The extent of crystallinity in LNG-IUSs was dependent on the crosslinking density of the polymer chains. The formulations with high crosslinking densities exhibited low degrees of crystallinity (Fig. 9C).

3.7. Polymer chain functionality using solid state ^{29}Si NMR

Variation in prepolymer ratios may affect the polymer functionality (arrangement of the polymer chain network around the crosslink junctions) which may subsequently impact the drug release of LNG-IUSs. The ^{29}Si CP/MAS NMR spectra of LNG-IUSs (Fig. 10) with different crosslinking densities showed a peak at approximately 25 ppm which could be ascribed to the PDMS monofunctional units. This may also be indicative of the presence of amorphous silica added to the prepolymers to increase the mechanical strength of PDMS (Birkefeld et al., 2003; Yue et al., 2013). It was observed that intensity of the peak at 25 ppm increased as the prepolymer B:A ratio increased. Signals with low intensities were observed at approximately -12 ppm which may be attributed to the difunctional units. Formulations with high PDMS crosslinking densities showed strong signals of the difunctional units ($-\text{O}-\text{Si}(\text{CH}_3)_2-$) at approximately -12 ppm (Beshah et al., 1986). The strong signals observed between -85 to -107 ppm confirm the presence of quaternary units in the crosslinked polymer. Differences in the intensities of quaternary units (Q2, Q3 and Q4) were observed for the formulations with different crosslinking densities which could also be attributed to amorphous silica component (Birkefeld et al., 2003). The formulations prepared using different mixing conditions (with a prepolymer B:A ratio of 1:1) showed minor differences in the ratio of the Q3 and Q4 units due to differences in their crosslinking densities (Supplementary data).

3.8. Microstructural changes in LNG-IUSs with different crosslinking densities

The drug-polymer reservoirs with different crosslinking densities showed significant differences in their topographical characteristics when observed under scanning electron microscopy (SEM) before and after drug release. As shown in Fig. 11A, the formulation with a prepolymer B:A ratio of 1:1 consists of solid drug particles dispersed in a solid-state matrix and was highly porous. On the other hand, the formulations with prepolymer B:A ratios of 0.5:1, 2:1, and 3:1 showed more gel-like yet porous polymer matrices inter-dispersed with precipitated polymer and drug. The formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, 0.33:1, 4:1 and 5:1 (lowest crosslinking densities) showed gel-like continuous polymer matrices with minimal or no porosity. The polymer matrix in these five formulations exists in the gel state coating the drug particles which are observed as bumps (Fig. 11A).

Fig. 11B shows that SEM images of all the formulations with different prepolymer ratios had relatively smoother matrices following complete drug release. The formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, 0.33:1 (low crosslinking densities) had completely smooth matrices with no pores. The formulation with a prepolymer ratio of 0.5:1 had a smooth polymer matrix with a few small pores. The formulations with prepolymer B:A ratios of 2:1, 3:1, 4:1 and 5:1 had relatively smooth matrices with multiple small pores. The formulation with a prepolymer B:A ratio of 1:1 had a relatively smoother matrix than prior to drug release. However, a few large polymer precipitates were observed together with multiple medium-sized pores.

Fig. 12 shows SEM images of the drug-polymer reservoirs (with a prepolymer B:A ratio of 1:1) prepared using different particle size and mixing conditions. Before drug release, all the formulations had solid drug particles dispersed in a solid-state matrix and were highly porous. The formulations prepared using the planetary speed mixer showed smaller pores than those prepared using manual mixing. The mean drug particle size in the polymer reservoirs prepared using the manual mixing method was approximately 7 μm and 2 μm for the non-micronized and micronized LNG, respectively. The mean drug particle size in the polymer reservoirs prepared using planetary speed mixing was reduced to approximately 4 μm and 1 μm for the non-micronized and micronized LNG, respectively. Following complete drug release, all the formulations with different particle size and mixing conditions had a relatively smooth matrix. The formulation with non-micronized LNG prepared using manual mixing had fewer polymer precipitates than before drug release. The pore size of this formulation was relatively larger than that of the other formulations. The formulation with micronized LNG prepared using manual mixing had a smooth polymer matrix with a few small pores. Formulations prepared using planetary speed mixing (micronized and non-micronized LNG) had smooth matrices with multiple small pores.

3.9. Effect of polymer swelling and porosity on release from LNG-IUSs

The swelling ability of drug-polymer reservoirs with different crosslinking densities were determined in *tert*-butanol, which was selected for this study since it was used in the accelerated release testing. The swelling ratio was calculated based on the differences in dimensions of the drug-polymer reservoir before and after solvent swelling. The formulations with prepolymer B:A ratios less than 1 showed lower swelling. It was observed that the formulation with a prepolymer B:A ratio of 1:1 (highest crosslinking density) demonstrated the highest swelling in *tert*-butanol. As prepolymer B:A ratios were further increased up to 5:1, swelling of the formulations decreased. The swelling ratio of all the formulations was correlated to their crosslinking densities as shown in Fig. 13A.

A mercury intrusion porosimeter was used to determine the porosities of the drug-polymer reservoirs with different crosslinking densities. It was observed that the porosity of the formulations increased as the crosslinking density was increased (Fig. 13B). The highest porosity was observed for the formulation with a prepolymer B:A ratio of 1:1. The formulations with higher swelling ratios and higher porosities showed faster drug release rates. The polymer swelling and porosity governed the release behavior of the LNG-IUSs as shown in Fig. 13C and 13D, respectively.

Formulations prepared using different drug particle size (with a constant prepolymer B:A ratio (1:1)) did not show significant differences in the porosity. The porosities and the swelling ratios of the formulations prepared using planetary speed mixing were higher than those prepared using manual mixing (Fig. 14A and 14B).

4. Discussion

4.1. Influence of prepolymer ratios and mixing conditions on the crosslinking density in LNG-IUSs

Crosslinking in PDMS is achieved through addition curing of the vinyl-terminated elastomer base and the hydrosilyl groups of the crosslinking agent in the presence of a platinum-based catalyst at high temperature. The ratio of prepolymers is critical for the formation of a crosslinked PDMS matrix and subsequently the release of the entrapped drug. Prepolymers used to form crosslinked PDMS are complex mixtures. Prepolymer A contains the PDMS elastomer base, a catalyst, hydrolysis products of silicones, and additives such as silica. Prepolymer B contains the elastomer base, a crosslinking agent, an inhibitor, hydrolysis products of silicones, and some additives. The PDMS crosslinking density increased as the ratio of prepolymers B:A was increased up to 1:1. This was attributed to the increasing amount of crosslinking agent in prepolymer B. The formulation with a prepolymer B:A ratio of 1:1 exhibited the highest crosslinking density (Fig. 4A). Further increase in the ratio of the prepolymers led to a decrease in the crosslinking density due to insufficient elastomer base and/or catalyst available for complete crosslinking (Fanse et al., 2021). A significant effect on the drug loading was also observed as the crosslinking densities of the formulations were varied. At prepolymer ratios other than 1:1, either prepolymer A or B is in excess. It is considered that some drug may bind to the excess prepolymer and therefore the drug may not be extracted completely into tetrahydrofuran leading to apparently lower drug loading (as shown in Table 1) (Murphy et al., 2016).

Another approach to prepare formulations with various crosslinking densities involved differences in the manufacturing process (*e.g.*, mixing conditions) of LNG-IUSs. The crosslinking densities of the formulations prepared using planetary speed mixing were higher than those prepared using manual mixing. This could be attributed to the higher interaction of the prepolymer chains with each other due to the dual centrifugal forces of the planetary speed mixer resulting in efficient mixing and thereby forming a higher number of crosslinked junctions. The drug loading of the formulations prepared using planetary speed mixing was lower than those prepared using manual mixing. This was considered to be due to the loss of drug particles through adhesion to the walls of the mixing vessel in the planetary mixer as evident from visual observation.

Although the crosslinking density in LNG-IUSs was sensitive to changes in prepolymer ratios and the mixing conditions during manufacturing, it was not significantly impacted by the drug particle size (Fig. 4C). This indicated that despite the differences in drug particle size and the high drug loading (50 % w/w), the prepolymer chains can establish sufficient contact with each other to form crosslinked junctions resulting in high crosslinking density.

4.2. Investigation of drug release mechanisms from LNG-IUSs based on the impact of crosslinking

According to conventional understanding, crosslinking in polymers forms a strong network structure capable of entrapping drug and causing a reduction in drug release rate (Martinez et al., 2014). For example, Teng *et al.* showed a decrease in the release rate of ciprofloxacin from crosslinked recombinant protein polymers (Teng et al., 2011). Similarly, Bukhari *et al.* reported that high crosslinking density in acrylic acid/gelatin hydrogels resulted in slow drug release rates (Bukhari et al., 2015). On the contrary, in this research, LNG-IUSs with high PDMS crosslinking density showed faster drug release compared to LNG-IUSs with lower crosslinking densities. It is essential to understand this unique behavior of highly crosslinked PDMS in such long-acting complex products. Accordingly, a comprehensive investigation of the effect of crosslinking density on the physicochemical properties of LNG-IUSs (including microstructural changes, swelling behavior, molecular weight between the crosslinked junctions, polymer functionality, and crystallinity) was carried out.

An optimum stoichiometric ratio of the PDMS elastomer base and the crosslinking agent (prepolymer B:A ratio of 1:1) is essential to achieve high crosslinking density in LNG-IUSs. Although the prepolymers A and B are initially in a gel state, at this prepolymer ratio, the chemically crosslinked PDMS has the highest molecular weight compared to the other prepolymer ratios and forms a solid-state matrix. Fig. 15 shows the microstructure on PDMS formation at different prepolymer B:A ratios. Chemical crosslinking occurs at specific junctions since the vinyl functional groups of the PDMS elastomer base react with the hydride functional groups of the crosslinking agent. This results in the formation of a branched network leaving large spaces between adjacent polymer chains leading to pockets or pores in the solid-state matrix (as observed in the SEM images). The results of mercury intrusion porosimetry confirmed that the formulations with high crosslinking densities had higher porosities than the formulations with low crosslinking densities. These differences in the porosities can also be explained from the results of the molecular weight of the polymer chains between the adjacent crosslink junctions (M_c). An inverse relationship exists between the M_c values and the PDMS crosslinking densities (Fig. 8B). The crosslinking density determines the number of crosslinks per unit volume (mol/cm^3). This means that more crosslink junctions are present per unit volume of PDMS at high crosslinking densities and thus the length of the polymer chain between the crosslinks is shorter compared to PDMS with low crosslinking densities (as shown in Fig. 16). These data explain why the formulations with high PDMS crosslinking densities had low M_c values. The molecular weight between the crosslinks affects the mesh size (ζ) of a polymer network which characterizes the free volume of the polymer available for diffusion (Carr and Peppas, 2009). Formulations with low M_c values form crosslinked PDMS networks with mesh sizes that facilitate mass transport resulting in faster drug release (Fig. 16).

In addition to the porosity/polymer mesh size, the polymer crystallinity in the LNG-IUSs was significantly affected due to PDMS crosslinking. Low crystallinity (%) was observed at high PDMS crosslinking densities (as evident from the X-ray scattering data) since the crosslink junctions act as constraints preventing the incorporation of neighboring polymer chains into the crystalline phase. These results are consistent with our previous findings

which revealed decreasing crystallization enthalpies of PDMS with increasing degrees of PDMS crosslinking (using DSC) (Fanse et al., 2021). PDMS is a semicrystalline polymer and the ratio of crystalline to amorphous domains in the polymer units are critical for drug release since the amorphous domains are permeable and accessible to the solvent molecules (O'dian, 1991). Since the formulation with high crosslinking density (prepolymer B:A ratio of 1:1) has a high porosity and more amorphous domains in the polymer chains, rapid organic solvent uptake (evident from the results of polymer swelling studies) occurred where the solvent was able to easily access the solid-state drug particles (Fig. 17). In addition, the drug molecules may exhibit higher solubility in the amorphous PDMS chains leading to faster drug diffusion. Consequently, this formulation has a faster drug release rate compared to the formulations with low PDMS crosslinking densities.

The polymers formed with an excess amount of either the PDMS elastomer base or the crosslinking agent remain in the gel state as they have low crosslinking densities with fewer site-specific chemically crosslinked junctions and therefore considerably lower porosity (Fig. 15). These formulations showed large variations in molecular weight (higher SD) indicating network inhomogeneity. The non-crosslinked polymer chains of PDMS form loop structures and free chain units as previously described (Di Lorenzo and Seiffert, 2015; Urayama et al., 2009). These physically entangled loop structures may result in higher apparent M_c values and lower effective mesh size (Fig. 16). Therefore, such a network structure may present a barrier to the diffusion of entrapped drugs from the polymer network. In addition, the polymer network of the formulations with low crosslinking densities had higher crystalline domains leading to lower solvent uptake. Additionally, highly lipophilic drugs such as LNG have a greater tendency to dissolve in the hydrophobic polymer and undergo drug diffusion through the polymer matrix. A lower PDMS crosslinking density may lead to smaller surface area of crosslink junctions per unit volume and the physically entangled polymer chains may provide a tortuous path leading to slower rates of drug diffusion through the matrix. Due to the highly aggregated dangling chains, low porosity and the need for the solvent to pass through the gel membrane of the relatively crystalline polymer to dissolve the drug, slower polymer swelling and subsequently slower drug release rates were observed for the formulations with low crosslinking densities (Figs. 16 and 17).

Specifically, the formulations with prepolymer B:A ratios of 0.5:1, 2:1, 3:1, and 4:1 had gel-like yet porous polymer matrices with precipitated polymer and drug. Whereas, the formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, 0.33:1, and 5:1 had very low crosslinking densities and thus consisted of more gel-like matrices with either very few or no pores. SEM images of all the formulations showed relatively smoother matrices following complete drug release which indicated that the polymer undergoes self-healing. The formulations with prepolymer B:A ratios of 0.2:1, 0.25:1, 0.33:1, and 0.5:1 are considered to undergo a larger extent of polymer self-healing since these formulations had smooth matrices with either very few or no pores after complete drug release. On the other hand, formulations with prepolymer B:A ratios of 2:1, 3:1, and 4:1 had multiple pores but with reduced pore size following complete drug release. Surprisingly, the formulation with a prepolymer B:A ratio of 5:1 had a more porous matrix than prior to drug release. This unusual behavior could be due to the high amount of crosslinking agent in this formulation

which prevented the healing of the pores following drug diffusion from the matrix. Another possible explanation for the reduced self-healing could be the high amount of silica fillers in the formulation with a prepolymer B:A ratio of 5:1 (as apparent from the results of ^{29}Si NMR).

The formulations with different prepolymer ratios demonstrated a non-linear increase in their polymer swelling ratios and porosities (Fig. 13C and D) which indicates more than one mechanism governing drug release from the LNG-IUSs depending on the release media. In addition to the previously discussed mechanisms, the variation in the amount of fillers with changes in the prepolymer ratios may also contribute to drug release from the LNG-IUSs. Fillers or additives such as silica are added to increase the mechanical strength of the formulations. The formulations with low crosslinking density (*e.g.*, prepolymer B:A ratio of 5:1) contain high silica fillers which may cause greater interactions mainly *via* the formation of hydrogen bonds. These hydrogen bonds may cause the strengthening of physically aggregated non-crosslinked polymer chains and thus contribute to their reduced swelling (Fig. 17).

Differences in the mixing conditions during the formulation of LNG-IUSs (investigated using the prepolymer B:A ratio of 1:1) also influenced drug release. The dual asymmetric centrifugal forces of the planetary speed mixing causes efficient mixing of the drug-polymer reservoirs and introduces air bubbles during mixing which may lead to higher porosity. This was confirmed by the SEM images of the formulations prepared *via* the planetary speed mixer which showed a larger number of small pores compared with the formulations prepared *via* manual mixing and further confirmed by the mercury intrusion porosimetry data. Higher porosity leads to higher solvent absorption and consequently faster drug release. The polymer swelling ratio data corroborates a release mechanism of swelling followed by diffusion-based drug release (Fig. 14).

Overall, the investigations conducted in the present study reveal that the physical state of the polymer (solid or gel state), molecular weight between the adjacent crosslinks, porosity and swelling-controlled drug diffusion, as well as polymer crystallinity are the various parameters significantly impacted by PDMS crosslinking which subsequently affect drug release. When the release media can penetrate the crosslinked PDMS polymer to allow polymer swelling, drug release from the formulations with high PDMS crosslinking density (the prepolymer B:A ratio of 1:1) is predominated by swelling followed by drug-diffusion. However, drug release from the formulations with low PDMS crosslinking densities is more complex due to excess amount of elastomer base, crosslinking agent, fillers, and silicone hydrolysis products. Accordingly, more than one of the aforementioned mechanisms (porosity-controlled swelling, gel membrane of the polymer acting as a barrier to solvent diffusion, crystallinity of polymer, interaction with excipients) may govern drug release from these formulations.

4.3. Selection of suitable model for release kinetics from LNG-IUSs

To better understand the structure-function relationship of polymeric systems and to elucidate the transport mechanisms, identifying a suitable mathematical model corresponding to the release behavior is critical. Since the design of LNG-IUSs involves

both the monolithic matrix and reservoir systems of controlled release, models such as the Higuchi, zero order, first order, and two-phase (zero order plus Higuchi) were investigated for data fitting. In addition to the above models, the Korsmeyer-Peppas model was also investigated. The Korsmeyer-Peppas model is a power-law model which is used to study drug release when the diffusion mechanism is unknown, or when more than one release mechanism is involved (Peppas and Narasimhan, 2014). Good data fitting was observed for the two-phase model, the Korsmeyer-Peppas model, and the first order model.

Analysis of the two-phase model showed that LNG-IUSs follow zero order release up to approximately 30% of drug release. This suggests that a constant activity source is maintained in the drug-polymer reservoir (up to 30% drug release) probably due to the excess drug particles in the solid-state providing a saturated condition for drug release. All the LNG-IUSs investigated followed the Higuchi model from approximately 30% to complete release. This suggests that drug diffusion is the dominating release mechanism following approximately 30% of drug release.

The Korsmeyer-Peppas model describes drug release in the range of 0–60% release. It was observed that PDMS swells in the presence of hydroalcoholic release media and the release properties may be dominated by the swelling kinetics of the polymer. The co-existence of two phenomena-diffusion and relaxation was established by fitting the data using the Korsmeyer-Peppas model (Tables 2A and 2B). The diffusion coefficient (n) is used in this model to characterize the drug transport mechanism. The LNG-IUSs drug-polymer reservoirs are cylindrical and hollow structures. Therefore, the interpretation of diffusion coefficient (n) was taken into account considering a planar geometry and based on the assumption that drug release occurs in one direction. For a release model from polymeric matrices with planar geometry, Fickian diffusion occurs when the coefficient (n) is 0.5 and non-Fickian anomalous transport occurs when $0.5 < n < 1$ (Peppas, 1985). The values of the coefficient (n) between 0.5 and 1 for all the LNG-IUS formulations (with different degrees of crosslinking) suggest drug release is governed by solvent diffusion in the matrix and swelling of the polymer. However, the different values of the coefficient (n) indicate that the rates of solvent diffusion and polymer swelling vary based on the differences in PDMS crosslinking.

Overall, the two-phase model explained the release mechanism of LNG-IUSs over their entire release profiles and the Korsmeyer-Peppas model explained the release profile up to 60% of drug release. However, all the LNG-IUSs investigated here followed the first order model for their entire release profiles (0–100% release) with high values for the regression coefficients and lower fluctuation in the rate constants with time compared to the other models. The first order model indicates that LNG-IUSs behave as a reservoir system through which the drug is released and the daily release rates decrease exponentially with time. The rate constants obtained using the first order model were used to establish correlations with PDMS crosslinking density and the physicochemical properties of LNG-IUSs (such as M_c , porosity, and swelling).

5. Conclusions

This research showed that PDMS prepolymer ratios and mixing conditions significantly affect the PDMS crosslinking density which subsequently impacts drug release from LNG-IUSs. The present study is the first report investigating drug release mechanisms from long-acting LNG-IUSs based on the impact of polymer crosslinking. It was concluded that the following factors influenced the drug release kinetics: porosity-controlled swelling, the amorphous/crystalline nature of the polymer, a diffusion barrier created by polymer coating on drug particles, as well as polymer-exciipient interactions which cause tightening of the non-crosslinked polymer chains leading to drug entrapment. Drug release from the LNG-IUSs with high PDMS crosslinking densities was primarily governed by porosity-controlled swelling and subsequent drug diffusion in release media containing *tert*-butanol. Whereas, release from the LNG-IUSs with low PDMS crosslinking densities was dominated by the other factors listed above. Furthermore, drug release from LNG-IUSs was independent of particle size indicating that drug diffusion through the polymer matrix was dominant over drug dissolution. Selection of suitable mixing conditions during the manufacturing of LNG-IUSs was shown to be critical with respect to drug release kinetics as the LNG-IUSs prepared using planetary mixing showed faster release rates as compared with LNG-IUSs prepared using manual mixing. A new accelerated release testing method was developed which significantly reduced the testing time for long-acting products such as LNG-IUSs. This will be useful for quality control purposes during formulation optimization and development. A correlation was established between the first order release rate constant and the PDMS crosslinking densities of LNG-IUSs. This will be helpful to predict drug release from LNG-IUSs with differences in PDMS crosslinking densities. The novel insights, based on thorough investigations carried out in this study, provide enhanced understanding of complex drug products such as LNG-IUSs. This will be useful in the development and assessment of LNG-IUSs as well as other PDMS-based controlled release systems including generic versions of these complex drugs.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations:

LNG	levonorgestrel
IUS	intrauterine system
IUD	intrauterine device
LNG-IUS	levonorgestrel intrauterine system
PDMS	polydimethylsiloxane

LARC	long-acting reversible contraceptives
IVIVC	<i>in vitro-in vivo</i> correlation
RLD	reference listed drug
Q1/Q2 equivalent	qualitatively and quantitatively equivalent
FDA	food and drug administration
DSC	differential scanning calorimetry
NMR	nuclear magnetic resonance
CP/MAS	cross polarization/magic angle spinning
WAXS	wide-angle X-ray scattering
SEM	scanning Electron microscopy
Tg	glass transition temperature

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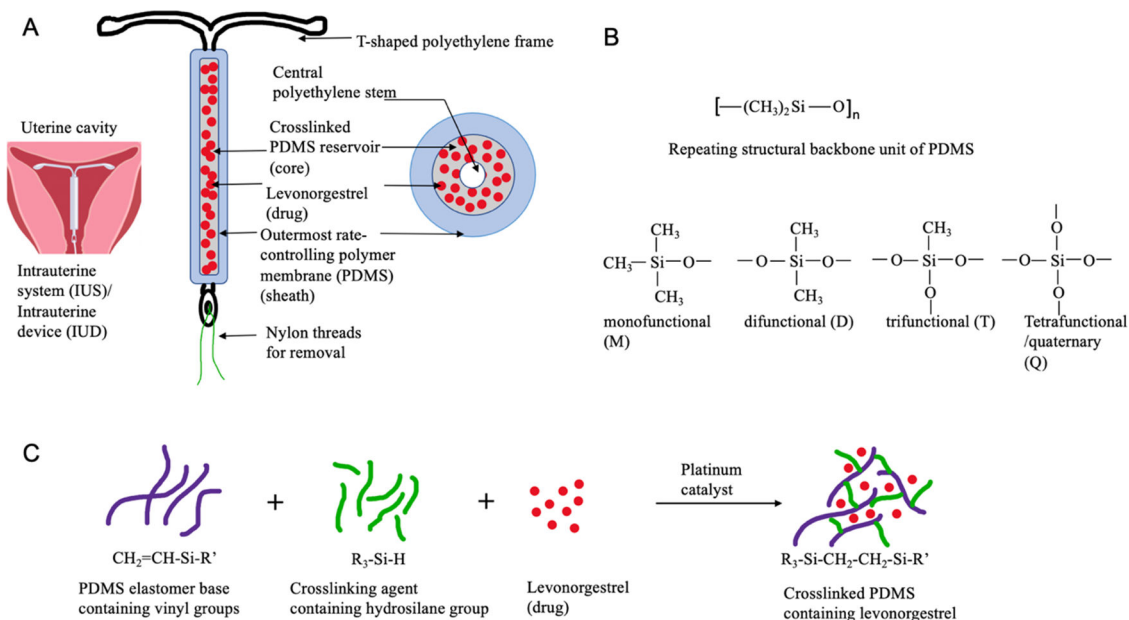
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**Fig. 1.**

A) The structure of levonorgestrel-intrauterine systems; B) various structural units of polydimethylsiloxane; and C) addition reaction resulting in a crosslinked PDMS matrix containing the drug.

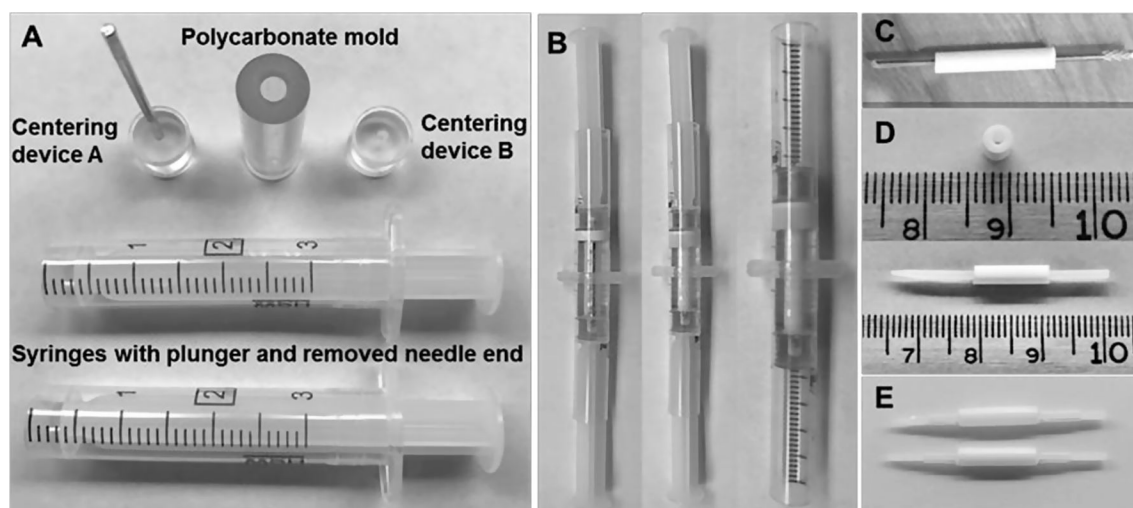
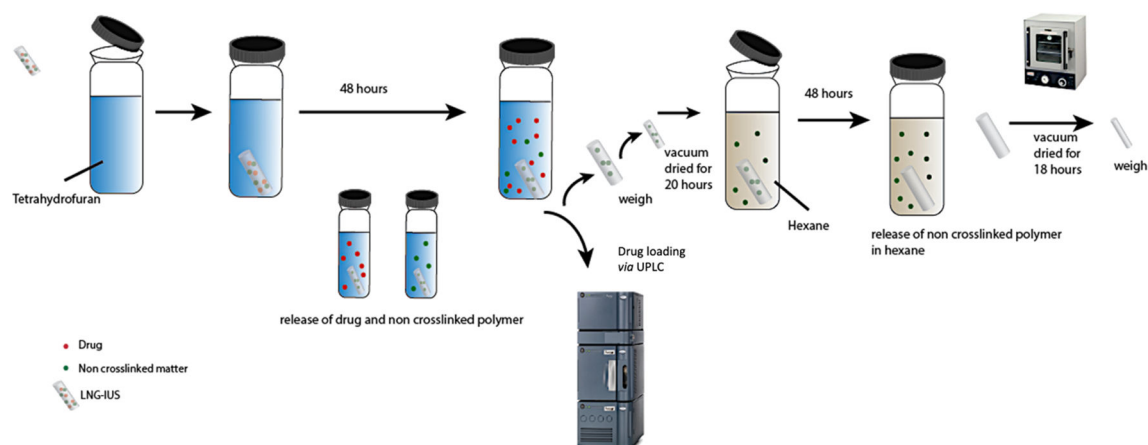


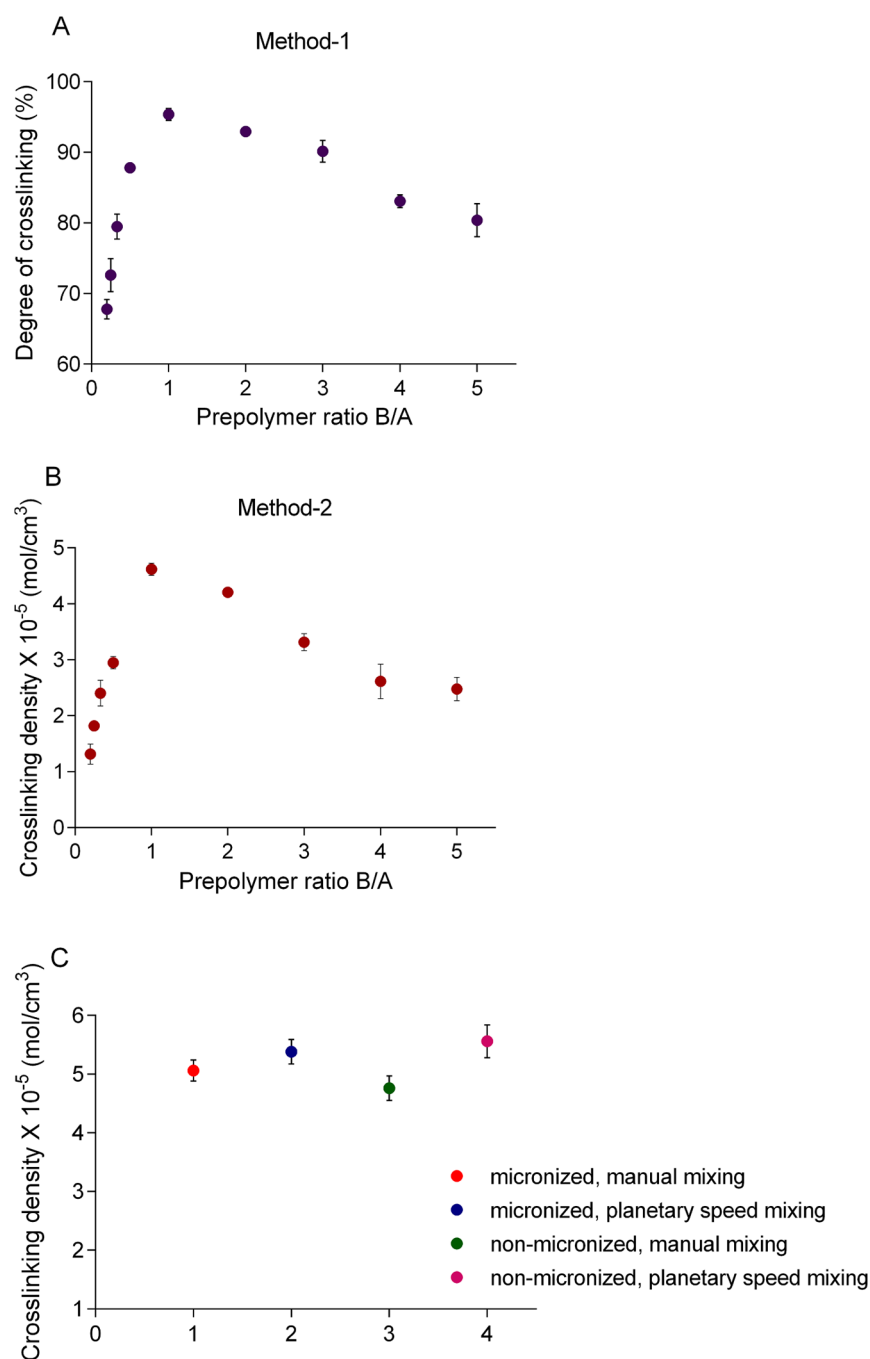
Fig. 2.
Pictorial representation of the preparation process of LNG-IUSs: A) Components of the customized extrusion device; B) The extrusion process of the PDMS polymer and the LNG mixture to form cylindrical drug-polymer reservoirs; C) The resultant drug-polymer reservoir with a steel rod; D) 100 mg reservoir mounted on a nylon rod; and E) LNG-IUS (with outer polymer membrane) (with permission from (Bao et al., 2018)).



Determination of PDMS crosslinking in LNG-IUSs via solvent extraction method

Fig. 3.

Pictorial representation of the solvent extraction method for determination of the degree of PDMS crosslinking in LNG-IUSs (with permission from (Fanse et al., 2021)).

**Fig. 4.**

A) Degree of PDMS crosslinking (%) in LNG-IUSs using solvent extraction (Method-I) (mean $\pm$ SD, n = 3); B) PDMS crosslinking densities (mol/cm³) of LNG-IUSs with different prepolymer ratios (Method-II) (mean $\pm$ SD, n = 3); and C) PDMS crosslinking densities (mol/cm³) of LNG-IUSs prepared using different drug particle sizes and mixing conditions (mean $\pm$ SD, n = 3).

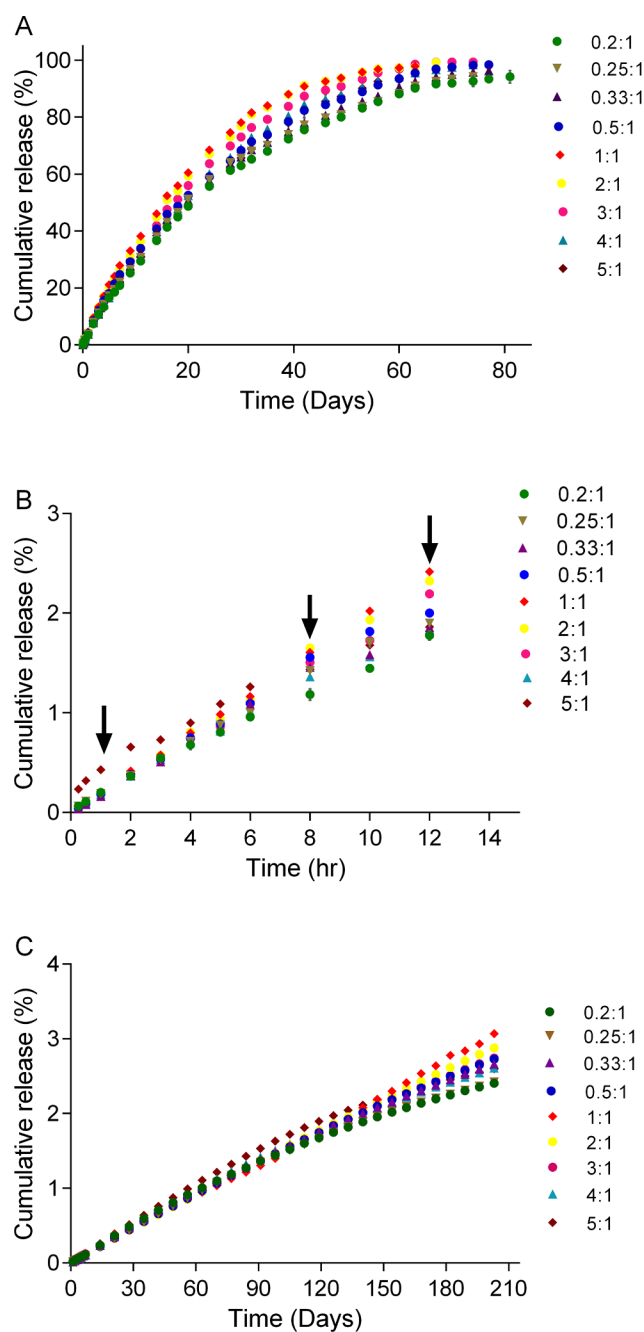


Fig. 5.

A) In vitro drug release profiles of LNG-IUSs with different prepolymer B:A ratios in accelerated release media containing 45% v/v *tert*-butanol in PBS at 65 °C in a water shaker bath at 100 rpm (n = 3); B) zoomed-in view of the accelerated release profiles from 0 to 12 h showing the initial burst release (n = 3); and C) real-time *in vitro* drug release profiles of LNG-IUSs with different prepolymer B:A ratios in release media containing 0.9% w/v saline at 37 °C (n = 3). (The error bars are so small that they are barely visible).

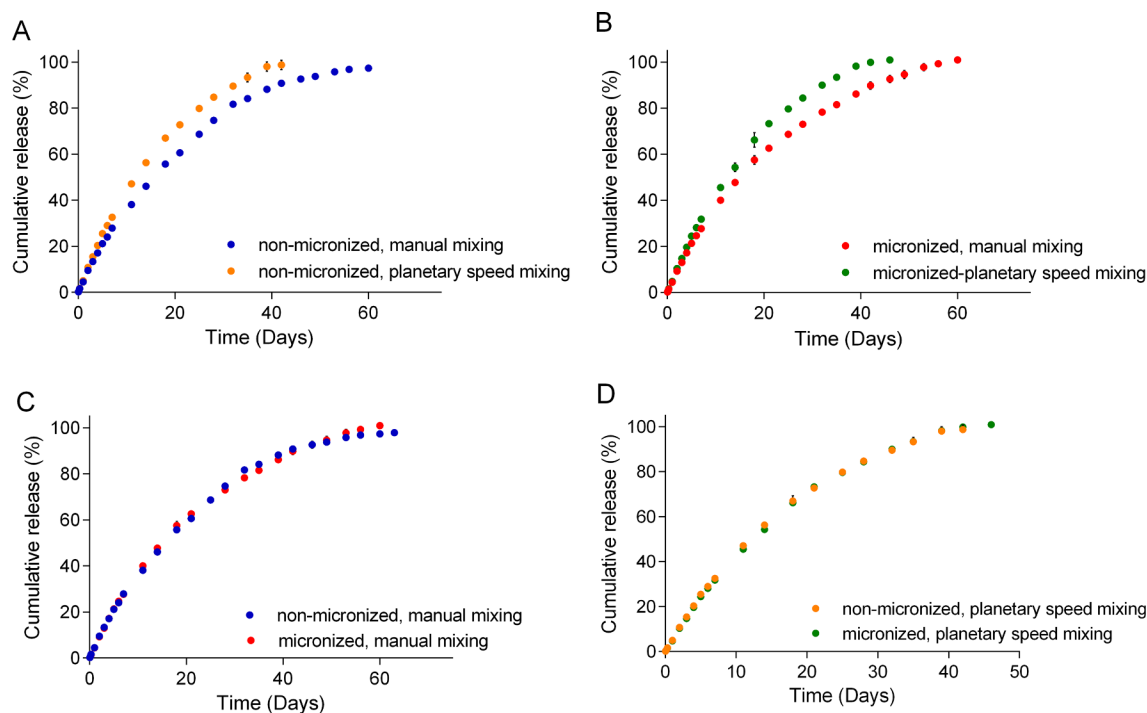
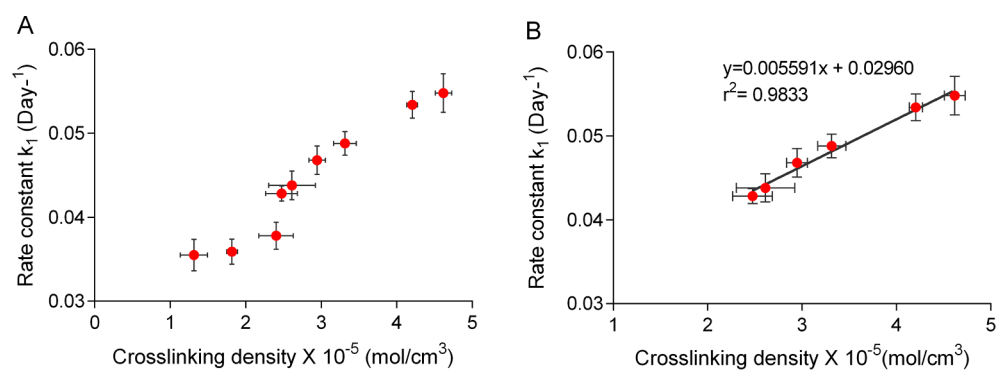


Fig. 6.

A) Accelerated *in vitro* release of LNG-IUSs with non-micronized drug prepared using different mixing methods (n = 3); B) accelerated *in vitro* release of LNG-IUSs with micronized drug prepared using different mixing methods (n = 3); C) accelerated *in vitro* release of LNG-IUSs with different particle size prepared using the manual mixing method (n = 3); and D) accelerated *in vitro* release of LNG-IUSs with different particle size prepared using the planetary mixing method (n = 3). (The error bars are so small that they are barely visible).

**Fig. 7.**

A) Correlation between the release rate constant and PDMS crosslinking density in LNG-IUSs; and B) linear regression of the release rate constant and PDMS crosslinking density.

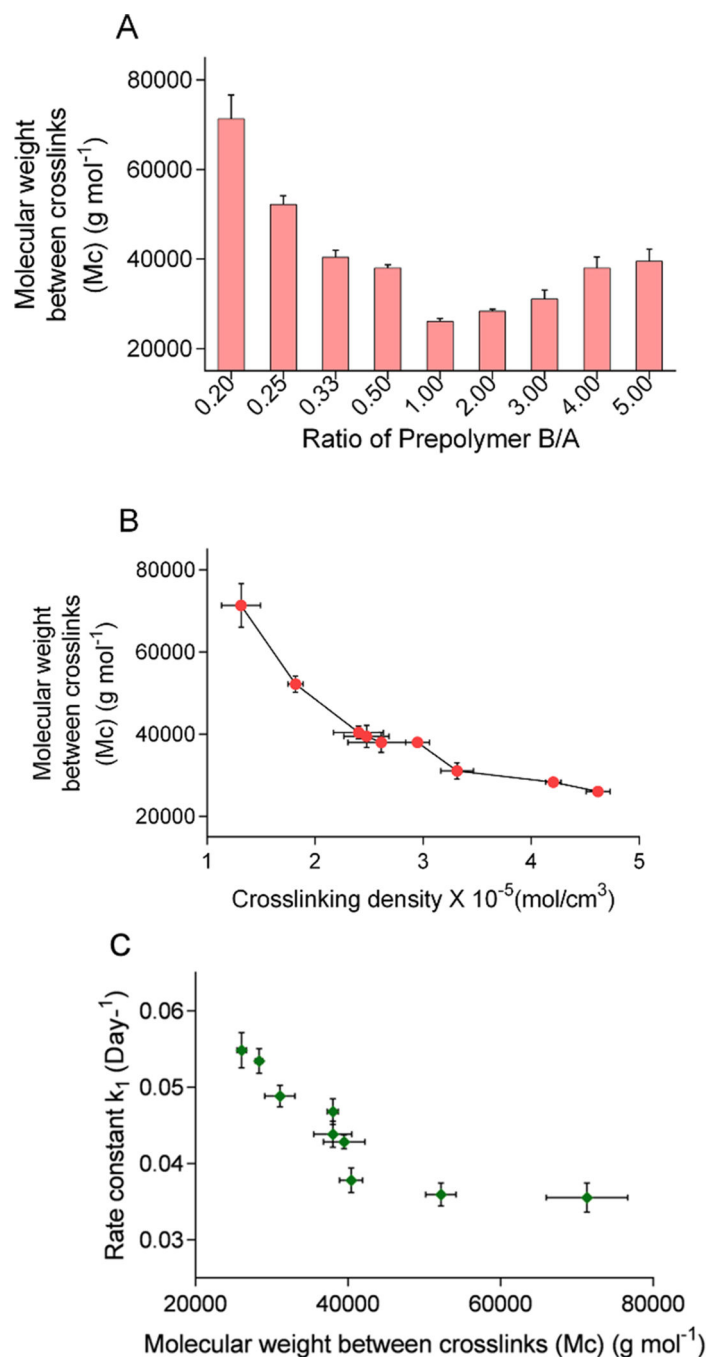
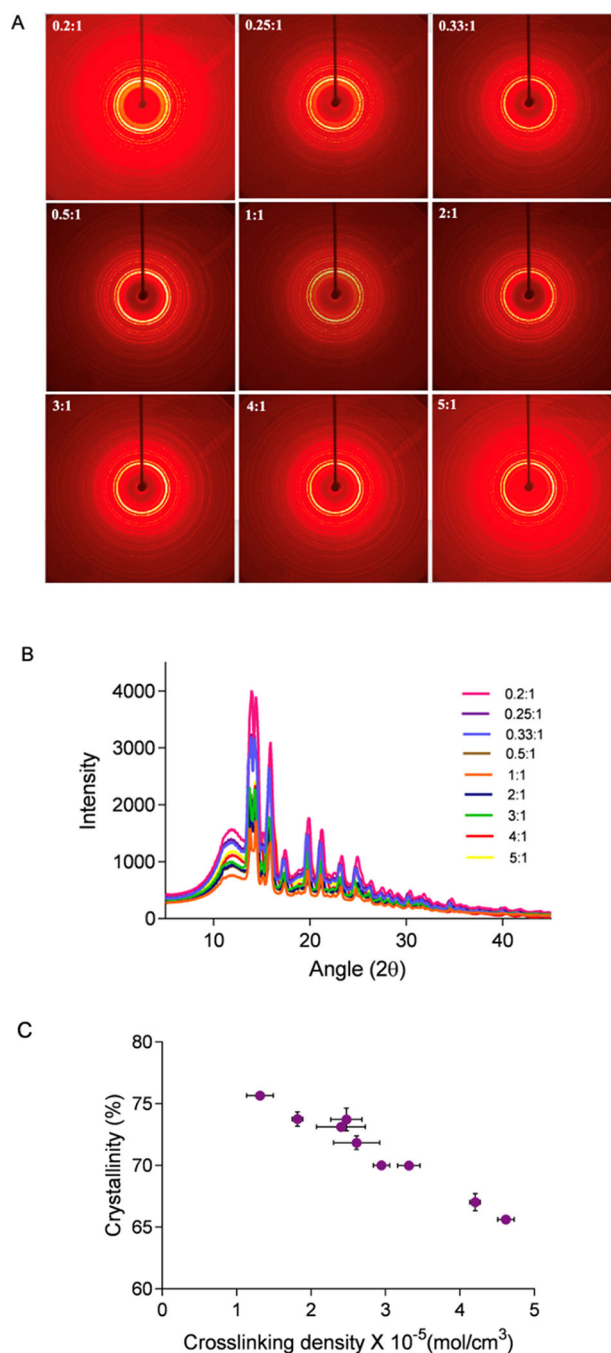


Fig. 8. Release behavior of LNG-IUSs is governed by the molecular weight between the crosslink junctions (Mc): A) changes in molecular weight between the crosslinks due to variation in the PDMS prepolymer ratios; B) molecular weight between the crosslinks as a function of the crosslinking densities of LNG-IUSs; and C) correlation between the first order release rate constant and the molecular weight between the crosslinks.

**Fig. 9.**

A) Representative 2D-WAXS frames for the formulations with different prepolymer B:A ratios; B) 1D-WAXS pattern of the formulations with different prepolymers B:A ratios indicating the crystalline and amorphous domains; and C) correlation between the crystallinity and PDMS crosslinking densities of LNG-IUSs ($n = 3$).

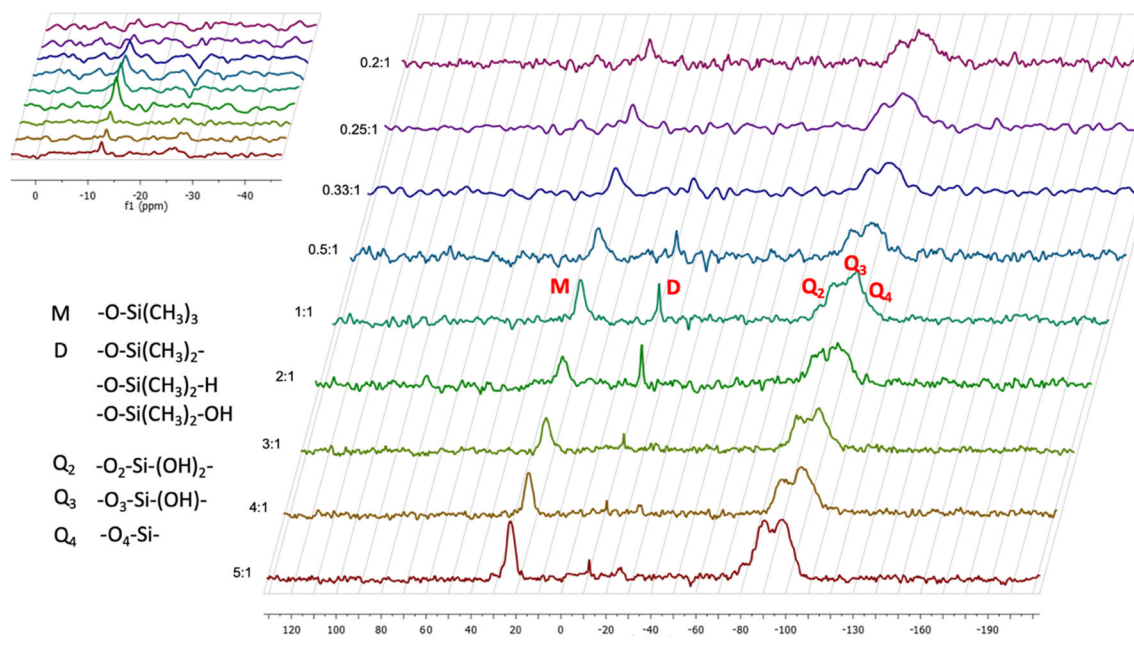


Fig. 10.
 ^{29}Si CP/MAS NMR spectra of LNG-IUSs with different crosslinking densities.

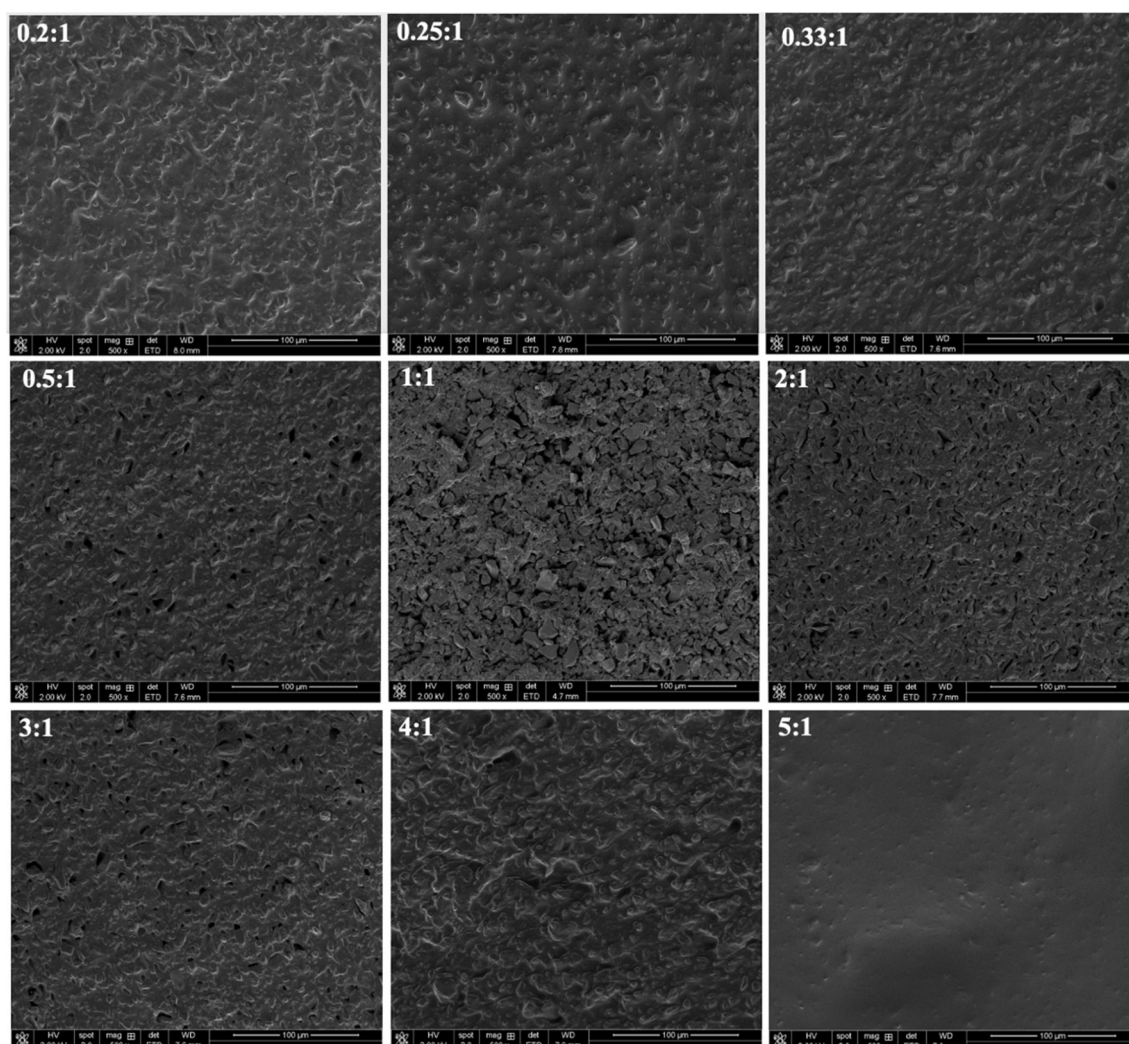


Fig. 11A.
SEM images (500X magnification) of the drug-polymer reservoirs of LNG-IUSs prepared using different ratios of prepolymers B:A before release testing.

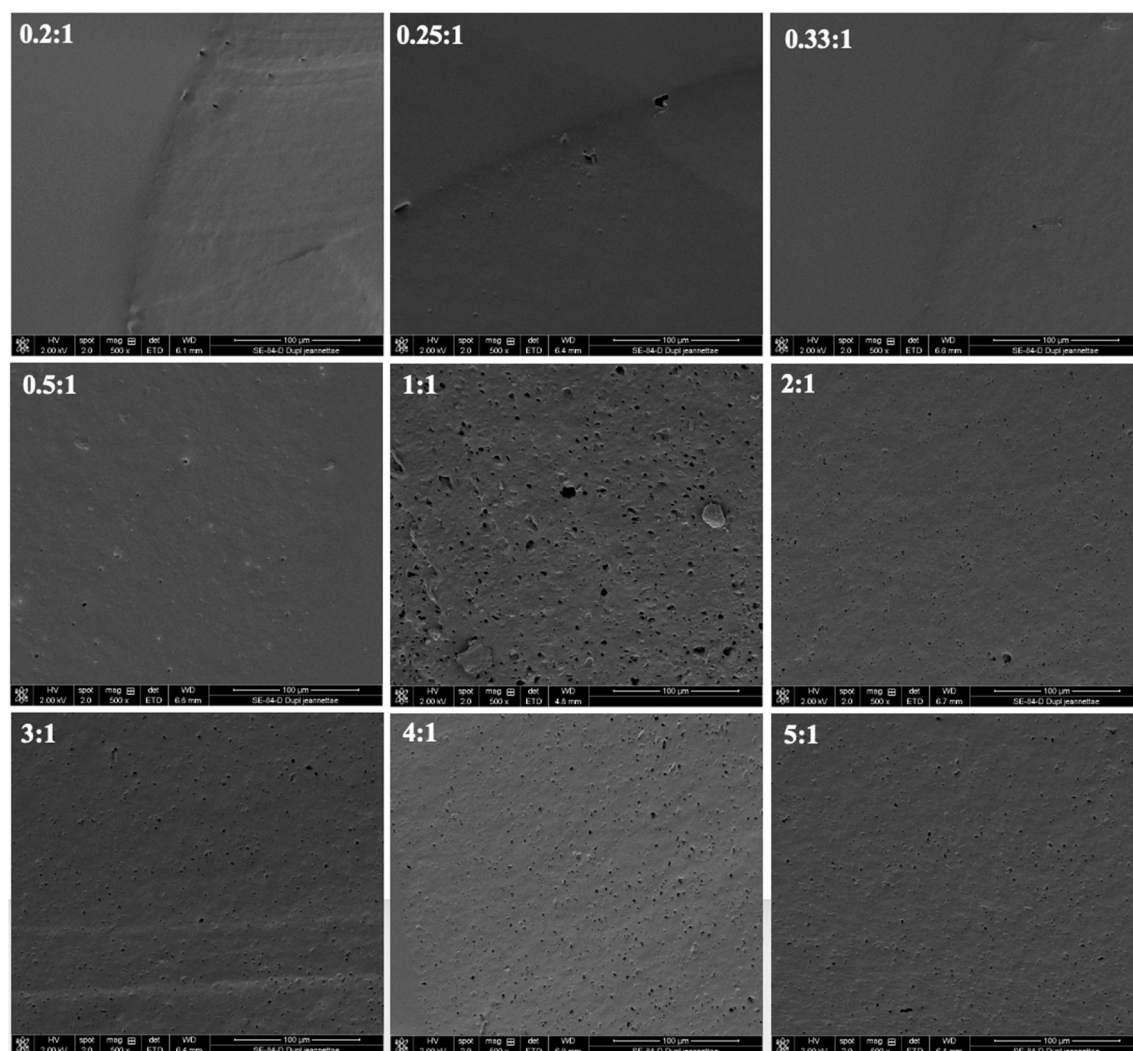


Fig. 11B.
SEM images (500X magnification) of the drug-polymer reservoirs of LNG-IUSs with different ratios of prepolymers B:A following complete drug release under accelerated conditions.

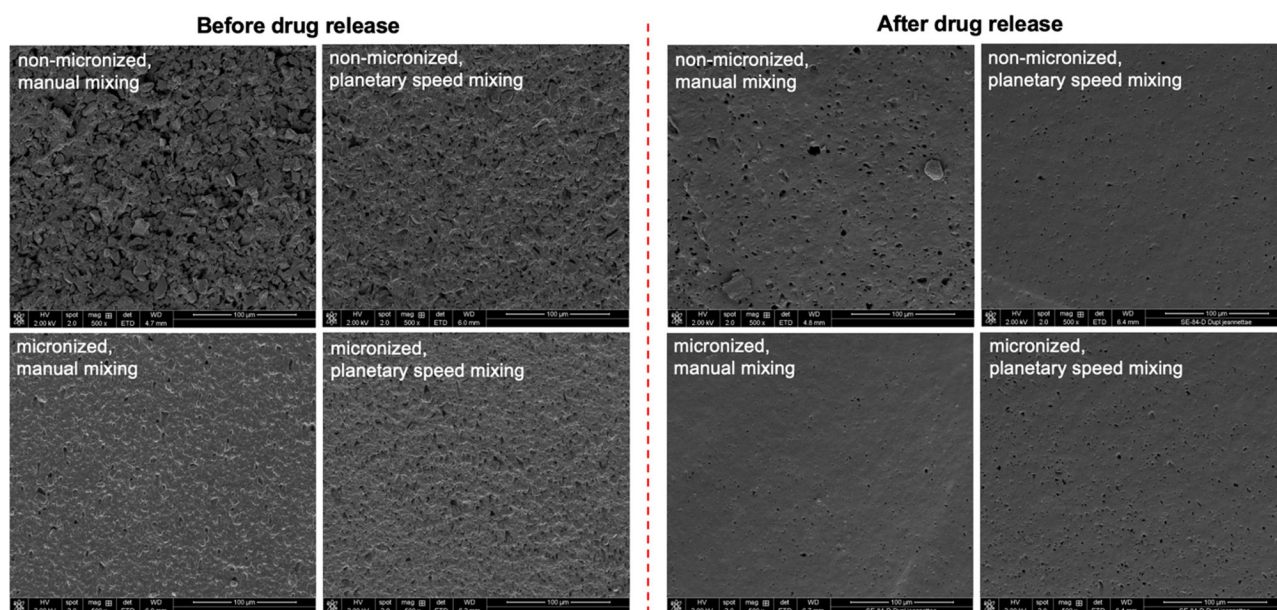
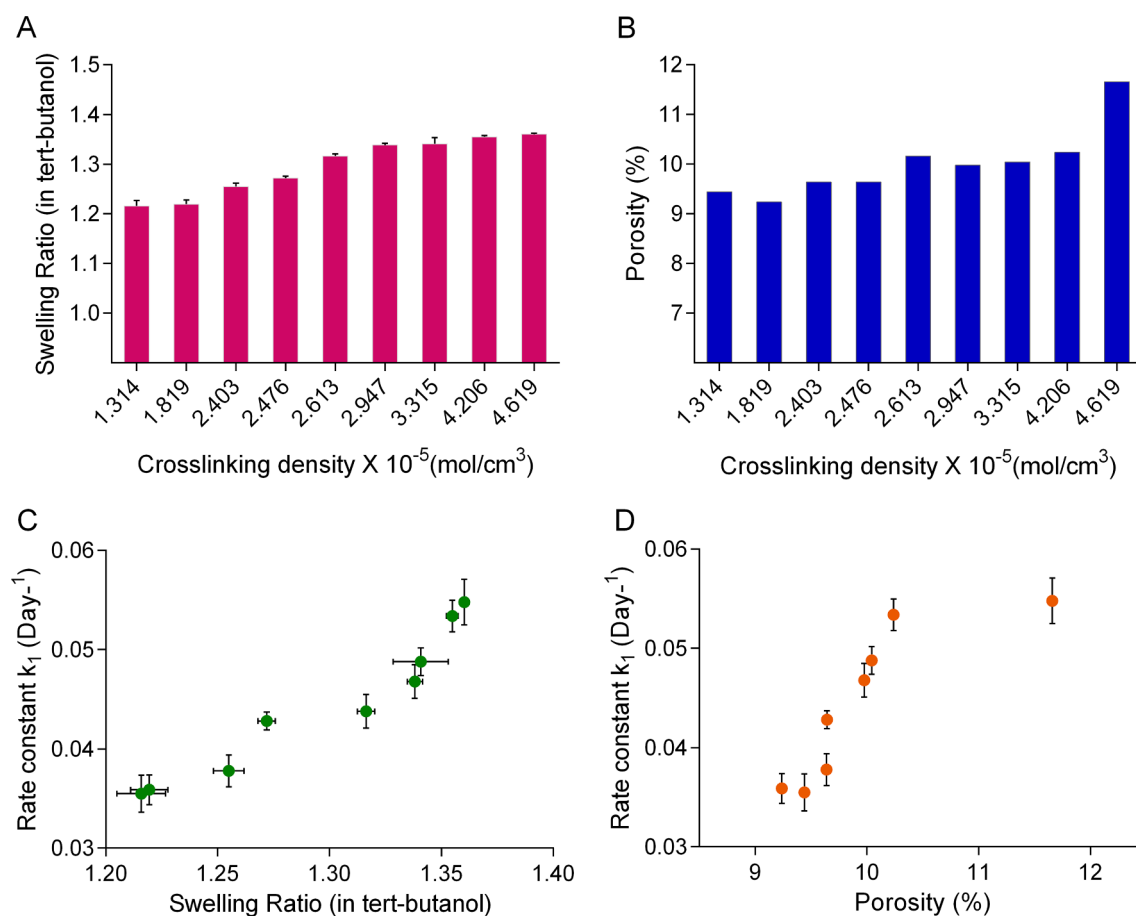


Fig. 12.
SEM images (500X magnification) of the drug-polymer reservoirs of LNG-IUSs with different particle size and mixing conditions (at a constant prepolymer B:A ratio of 1:1) before and after drug release.

**Fig. 13.**

A) Swelling ratio in *tert*-butanol as a function of the crosslinking densities in LNG-IUSs (n = 3); B) differences in the porosities of LNG-IUSs with varying crosslinking densities; C) relationship between release rate constant and swelling ratio of LNG-IUSs (n = 3); and D) relationship between release rate constant and porosity in LNG-IUSs (n = 3).

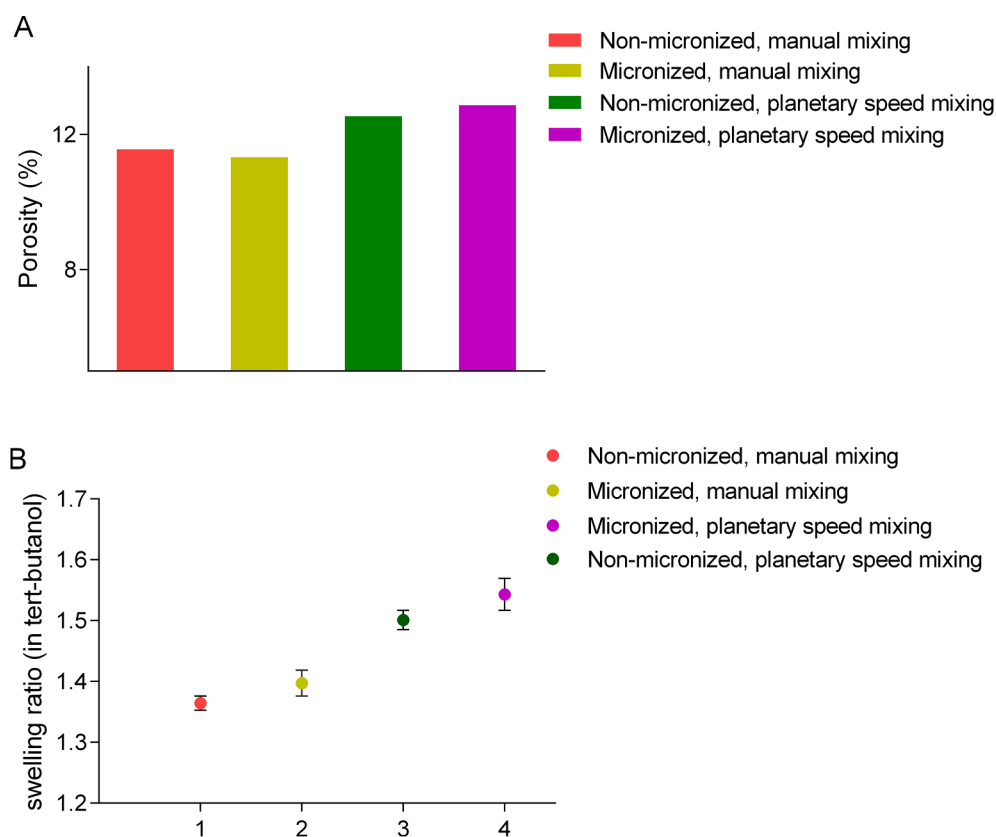


Fig. 14.
A) Porosity of LNG-IUSs prepared using different drug particle size and mixing conditions; and B) swelling ratio of drug-polymer reservoirs (in *tert*-butanol) of LNG-IUSs prepared using different drug particle size and mixing conditions (n = 3).

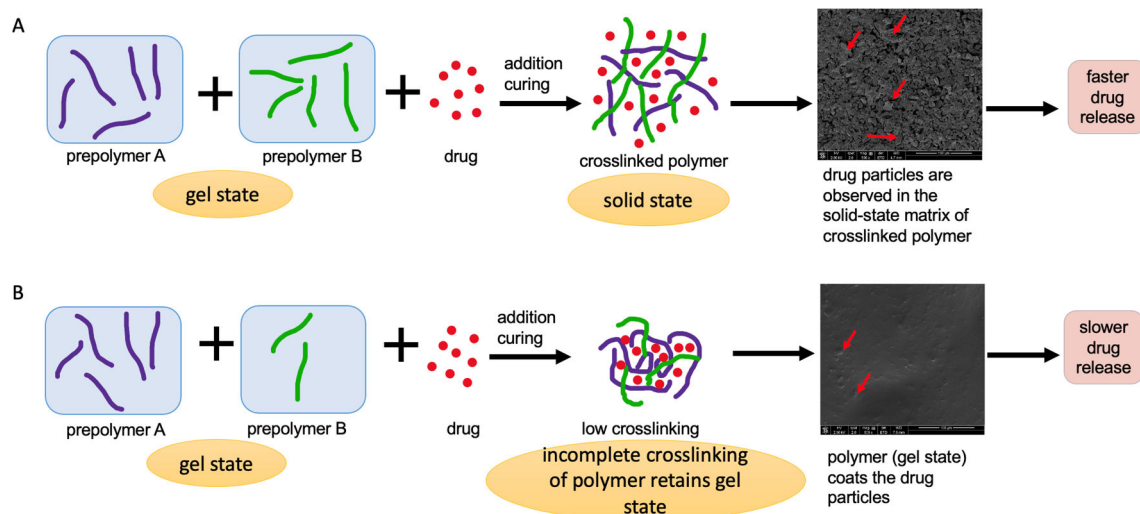


Fig. 15. Microstructural changes in LNG-IUSs with different prepolymer ratios resulting in different drug release rates.

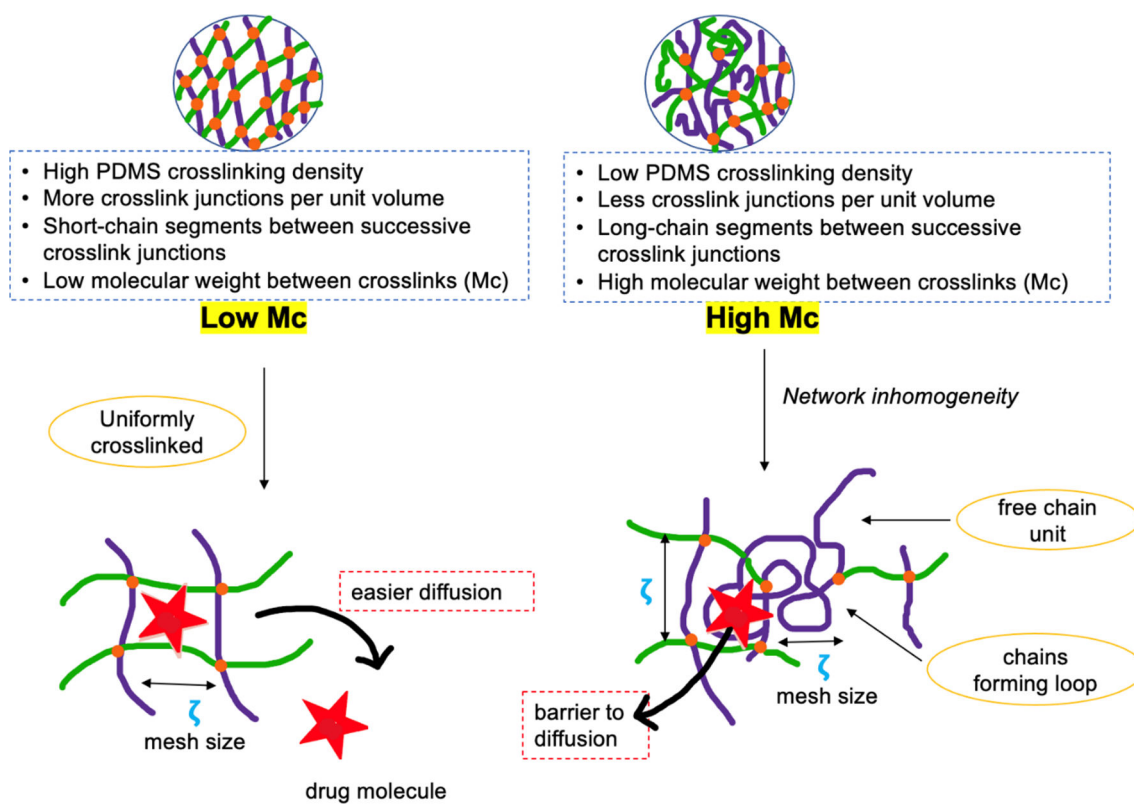
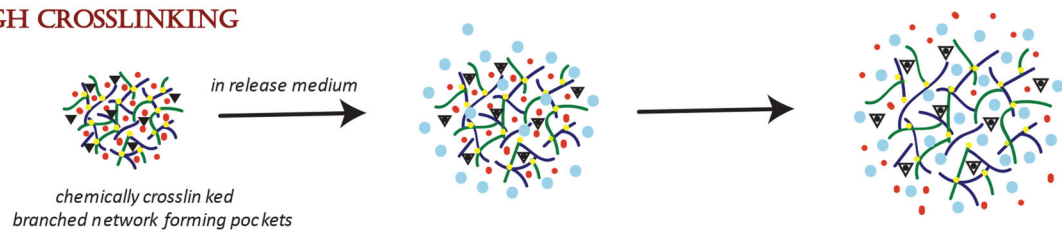


Fig. 16.

Effect of molecular weight between the PDMS crosslink junctions (M_c) on the effective mesh size and drug diffusion from the polymer matrix.

HIGH CROSSLINKING



LOW CROSSLINKING



Fig. 17.
Swelling and diffusion-controlled drug release of LNG-IUSs with different crosslinking densities.

Table 1

The drug loading of LNG-IUS drug-polymer reservoirs prepared using different prepolymer ratios, drug particle sizes and mixing conditions (n = 3).

LNG-IUS	Observed drug loading (% w/w)	RSD (%)
Different prepolymer ratios		
0.2:1	44.52	2.09
0.25:1	43.90	2.18
0.33:1	44.76	1.45
0.5:1	46.71	1.02
1:1	48.74	1.61
2:1	46.41	1.02
3:1	44.47	1.27
4:1	43.70	1.69
5:1	44.21	1.11
Different drug particle size and mixing technique (with a prepolymer B:A ratio of 1:1)		
micronized LNG-manual mixing	47.81	1.59
micronized LNG-planetary speed mixing	45.74	0.78
non-micronized LNG-manual mixing	48.74	1.61
non-micronized LNG-planetary speed mixing	46.18	1.51

Table 2A

Model fitting parameters for the release profiles of LNG-IUSs with different prepolymer ratios.

Model	First order model ^a			Two-phase model ^a			Higuchi model			Korsmeyer-Peppas model ^b		
	Equation	Q (%) = 100 (1-e ^{-k₁t})	r ²	T _{50%} (day)	Zero order model	Q (%) = k ₀ t 0 to 30% release	Q (%) = k _H t ^{1/2} 30% to complete release	r ²	k _H	Q (%) = k _p t ⁿ	k _p	r ²
Ratio of prepolymer B:A												
		k ₁ X 10 ⁻²			k ₀	r ²	k _H	r ²				n
0.2:1		3.55 ± 0.01	0.9970 ± 0.0011	19.54	2.74 ± 0.01	0.9871 ± 0.0008	12.15 ± 0.02	0.9905 ± 0.0007		3.38 ± 0.02	0.9989 ± 0.0005	0.85
0.25:1		3.59 ± 0.02	0.9950 ± 0.0008	19.29	2.83 ± 0.01	0.9869 ± 0.0005	12.06 ± 0.02	0.9822 ± 0.0012		3.44 ± 0.00	0.9988 ± 0.0003	0.90
0.33:1		3.78 ± 0.04	0.9914 ± 0.0004	18.35	3.17 ± 0.07	0.9935 ± 0.0013	12.96 ± 0.02	0.9894 ± 0.0015		3.51 ± 0.01	0.9971 ± 0.0001	0.91
0.5:1		4.68 ± 0.07	0.9964 ± 0.0007	14.82	3.34 ± 0.01	0.9917 ± 0.0001	13.87 ± 0.03	0.9911 ± 0.0005		3.73 ± 0.02	0.9969 ± 0.0003	0.93
1:1		5.48 ± 0.03	0.9916 ± 0.0008	12.64	4.00 ± 0.02	0.9968 ± 0.0004	17.32 ± 0.03	0.9951 ± 0.0008		4.14 ± 0.01	0.9966 ± 0.0005	0.96
2:1		5.34 ± 0.05	0.9945 ± 0.0009	12.97	3.80 ± 0.01	0.9946 ± 0.0003	17.47 ± 0.02	0.9954 ± 0.0006		4.09 ± 0.04	0.9977 ± 0.0012	0.95
3:1		4.88 ± 0.03	0.9929 ± 0.0011	14.19	3.37 ± 0.02	0.9937 ± 0.0005	15.83 ± 0.01	0.9880 ± 0.0017		4.05 ± 0.00	0.9991 ± 0.0002	0.92
4:1		4.38 ± 0.05	0.9953 ± 0.0007	15.84	2.90 ± 0.06	0.9899 ± 0.0012	14.92 ± 0.02	0.9852 ± 0.0022		3.92 ± 0.03	0.9992 ± 0.0007	0.91
5:1		4.28 ± 0.04	0.9884 ± 0.0013	16.18	2.85 ± 0.02	0.9901 ± 0.0006	14.73 ± 0.01	0.9905 ± 0.0014		3.60 ± 0.04	0.9970 ± 0.0013	0.90

Q indicates the cumulative release percentage and k represents the rate constant.

^aThe fitting was made for the full drug release profile.

^bThe fitting was made up to 60% of drug release from LNG-IUSs.

Table 2B

Model fitting parameters for the release profiles of LNG-IUSs (with prepolymer B:A ratio of 1:1) prepared using different particle size and mixing conditions.

Model	First order model ^a			Two-phase model ^d			Korsmeyer-Peppas model ^b		
Equation	Q (%) = 100 (1-e ^{-k₁t})			Zero order model			Higuchi model		
	k ₁ X 10 ⁻²	r ²	T _{50%} (day)	Q (%) = k ₀ t 0 to 30% release		Q (%) = k _H t ^{1/2} 30% to complete release	k _p	r ²	n
				k ₀	r ²				
micronized, manual mixing	5.41 ± 0.07	0.9937 ± 0.0012	12.81	4.06 ± 0.02	0.9972 ± 0.0004	14.47 ± 0.05	0.9922 ± 0.0007	4.19 ± 0.04	0.9977 ± 0.0006
micronized, planetary mixing	6.75 ± 0.08	0.9909 ± 0.0008	10.27	4.79 ± 0.04	0.9986 ± 0.0005	18.57 ± 0.08	0.9914 ± 0.0012	4.68 ± 0.04	0.9719 ± 0.0008
non-micronized, manual mixing	5.48 ± 0.03	0.9913 ± 0.0005	12.49	4.00 ± 0.02	0.9968 ± 0.0004	17.32 ± 0.04	0.9951 ± 0.0008	4.14 ± 0.02	0.9966 ± 0.0005
non-micronized, planetary mixing	6.72 ± 0.06	0.9939 ± 0.0007	10.31	4.96 ± 0.03	0.9983 ± 0.0004	18.47 ± 0.07	0.9922 ± 0.0011	4.78 ± 0.05	0.9987 ± 0.0007

Q indicates the cumulative release percentage and k represents the rate constant.

^aThe fitting was made for the full drug release profile.

^bThe fitting was made up to 60% of drug release from LNG-IUSs.